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Mouse sensorimotor cortex reflects complex kinematic details during reaching and grasping

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The granularity with which neural activity in the sensorimotor cortex of mice corresponds to voluntary forelimb motion is a key open question. This paper provides **compelling** evidence for the encoding of low-level features like joint angles and represents an **important** step forward toward understanding cortical limb control signals.

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Abstract

Coordinated forelimb actions, such as reaching and grasping, rely on motor commands that span a spectrum from abstract target specification to detailed instantaneous muscle control. The sensorimotor cortex is central to controlling these complex movements, yet how the detailed command signals are distributed across its numerous subregions remains unclear. In particular, in mice it is unknown if the primary motor (M1) and somatosensory (S1) cortices represent low-level joint angle details in addition to high-level signals like movement direction. Here, we combine high quality markerless tracking and two-photon imaging during a reach-to-grasp task to quantify movement-related activity in the mouse forelimb M1 (M1-fl) and forelimb S1 (S1-fl). Linear decoding models reveal a strong representation of proximal and distal joint angles in both areas, and both areas support joint angle decoding with comparable fidelity. Despite shared low-level encoding, the time course of high-level target-specific information varied across areas. M1-fl exhibited early onset and sustained encoding of target-specific signals while S1-fl was more transiently modulated around lift onset. These results reveal both shared and unique contributions of M1-fl and S1-fl to reaching and grasping, implicating a more distributed cortical circuit for mouse forelimb control than has been previously considered.

Introduction

Primary motor cortex (M1) and primary somatosensory cortex (S1) are the neocortical structures most strongly and directly connected to the sensorimotor periphery. Accordingly, signals related to movement are strongly present in M1 (Evarts 1973 [↗](#); Galiñanes et al. 2018 [↗](#); Georgopoulos et al. 1982 [↗](#); Moran and Schwartz 1999 [↗](#); Park et al. 2022 [↗](#); Saleh et al. 2012 [↗](#); Sauerbrei et al. 2020 [↗](#)), and signals related to both cutaneous and proprioceptive somatosensation are strongly present in S1 (Alonso et al. 2023 [↗](#); Chowdhury et al. 2020 [↗](#); Costanzo and Gardner 1981 [↗](#); Fromm and Evarts 1982a [↗](#); Gardner and Costanzo 1981 [↗](#); Pei et al. 2009 [↗](#); Prsa et al. 2019 [↗](#); Umeda et al. 2019 [↗](#)). Yet despite being among the longest-studied brain structures, our understanding of these areas' exact roles in movement production remains incomplete (Omrani et al. 2017 [↗](#)).

In both primates and rodents, M1 and S1 are known to be strongly interconnected (Pons and Kaas 1986; Huerta and Pons 1990; Darian-Smith et al. 1993; Yamawaki et al. 2021; Muñoz-Castañeda et al. 2021) and work closely together (Nelson 1996): sensory signals are often strongly present in M1 (Lucier et al. 1975; Lemon et al. 1976; Evarts and Fromm 1977; Wong et al. 1978; Fetz et al. 1980; Fromm et al. 1984; Alonso et al. 2023; Asanuma 1975), and movement related signals are often present in S1 (Chowdhury et al. 2020; Cohen et al. 1994; Goodman et al. 2019; Nelson 1987; O'Connor et al. 2021; Prud'homme and Kalaska 1994; Tanji and Wise 1981; Umeda et al. 2019; Weber et al. 2011). Nevertheless, stimulation in M1 produces movement much more readily than in S1 (Penfield and Boldrey 1937; Tennant et al. 2011), and lesions (Lawrence and Kuypers 1968; Whishaw et al. 1993; Kawai et al. 2015; Mizes et al. 2023, 2024; Nicholas and Yttri 2024) and inactivations (Galiñanes et al. 2018; Guo et al. 2015; Morandell and Huber 2017; Miri et al. 2017; Hwang et al. 2019; Sauerbrei et al. 2020; Hwang et al. 2021; Park et al. 2022) of M1 produce strong deficits in movement production, especially for movements that are kinematically complex (Bollu et al. 2024), flexible (Heindorf et al. 2018; Mizes et al. 2023), or coordinated across the hand and arm (Guo et al. 2015).

In monkeys, motor cortical activity has been argued to relate to high-level signals, such as movement endpoint (Georgopoulos et al. 1986) or hand velocity (Kalaska et al. 1983; Ashe and Georgopoulos 1994; Paninski et al. 2004; Wang et al. 2007); or, low-level signals such as joint kinematics (Vargas-Irwin et al. 2010; Saleh et al. 2012, 2010; Goodman et al. 2019; Okorokova et al. 2020) or muscle forces (Kakei et al. 1999; Lemon et al. 1986; Morrow and Miller 2003); or even very detailed control previously thought to be reserved for the spinal cord, such as selecting types of motor pools (Marshall et al. 2022). The presence of signals at all of these levels suggests that monkey M1 may play a role in multiple levels of control. However, it is challenging to tease these roles apart in a model system where fine-grained causal perturbations and circuit tools are less developed.

The mouse has emerged recently as a model of interest for dissecting motor control (Warriner et al. 2020). As a model system, it offers high throughput, detailed atlases through both anatomical and functional tracing (BRAIN Initiative Cell Census Network (BICCN) et al. 2021; Muñoz-Castañeda et al. 2021; Winnubst et al. 2019; Zingg et al. 2014), and powerful techniques for large-scale recording (Manley et al. 2024; Steinmetz et al. 2021; Yu et al. 2021) and perturbation (Guo et al. 2015, 2021; Sauerbrei et al. 2020). However, the study of systems-level motor control in actively behaving animals is less mature in the mouse than in primates.

Extensive motor mapping experiments in rodents have revealed that activating different parts of the sensorimotor cortex evokes movements of different body parts or different kinds of movements of the same body part, as it does in primates (for review, see (Harrison and Murphy 2014)). Yet it is unclear how the topography of stimulation-evoked movements relates to the roles of these areas during volitional actions. Perturbations during behavioral tasks in mice involving forelimb lever or reaching movements have provided a coarse-level understanding of how these areas contribute during behavior. Inactivations and lesions of M1 and S1 have shown that M1 is required for the execution of dexterous reach-to-grasp movements (Guo et al. 2015; Sauerbrei et al. 2020; Galiñanes et al. 2018; Wang et al. 2017; Whishaw et al. 1991; Whishaw 2000) and that S1 is essential for adapting learned movements to external perturbations of a joystick (Mathis et al. 2017). However, spinal cord projections from mouse M1 and S1 primarily target spinal interneurons rather than directly synapsing onto motor neurons (Gu et al. 2017; Wang et al. 2017; Ueno et al. 2018), suggesting cortical activity might play a more modulatory role. Further, stimulation of brainstem nuclei alone can evoke naturalistic forelimb actions, including realistic reaching movements involving coordinated flexion and extension of the proximal and distal limb (Esposito et al. 2014; Ruder et al. 2021; Yang et al. 2023). Taken together, these results have raised the question of what role mouse M1 and S1 play in the control of goal-directed forelimb movements.

One route to answering this question involves characterizing the signals present in mouse M1 and S1 during movement. If mouse M1 and S1 were to control only high-level aspects of forelimb movements, activity should be dominated by ‘abstract’ signals like target location and reflect little trial-to-trial variability in reach kinematics. If instead M1 and S1 control low-level movement features then activity should correlate strongly with forelimb joint angle kinematics and their trial-to-trial variation when reaching to different targets. While the presence of high- or low-level signals in a cortical area does not necessarily imply that they are causally responsible for these aspects of movement, characterizing what signals are present serves as a first step toward determining how these areas relate to movement.

Here we take this step by characterizing what kinematic information is present in M1 and S1 during reaching and grasping. To do so we combine a reach-to-grasp-to-drink paradigm (based on (Galiñanes et al. 2018 [↗](#); Galinanes and Huber 2023 [↗](#))) with detailed high-speed behavior tracking and two-photon calcium imaging to ask what movement-related signals are present in mouse M1 and S1. Decoding models revealed that both M1 and S1 encode fine details of movement in the mouse, enabling decoding of all 24 joint angles we reconstructed from the shoulder to digits – even features such as digit splay and opposition within the paw. Differences between the areas emerged only in the decoding of higher-level aspects of the task: M1 reflected target location earlier than S1 and S1 reflected movement initiation time more accurately than M1. Our results link mouse M1 and S1 activity to the execution of reach-to-grasp movements and highlight their distinct yet complementary roles in encoding low-level kinematic details and higher-order movement features.

Results

Complex movement kinematics during mouse reaching and grasping movements

Assessing the relationship between neural activity and the details of movement requires striking a balance between achieving repeatable behavior and evoking sufficient trial-to-trial variability to broadly sample movement space. To achieve this balance, we modified a previously-developed water-reaching task (Galiñanes et al. 2018 [↗](#)) to evoke challenging reaching and grasping movements to two locations (Fig. 1A [↗](#)). Importantly, the task setup was made compatible with high-speed stereo camera imaging and modern markerless tracking techniques for tracking putative joint centers, allowing for a detailed quantification of reaching trajectories and grasp attempts. Using a custom 3D-aware frame-labeling GUI and large numbers of labeled frames, we achieved high-quality tracking on nearly every frame (99% of lift-locked frames passed quality assurance for all 24 joint angles; 94.5% of trials were tracked usably; see Methods). This allowed us to quantify the kinematics at a fine level of detail and to obtain a more comprehensive description of these movements through joint angle kinematics (see Fig. 3 [↗](#)).

Our task involved two water spout targets fixed in place on either side of the mouse’s head at distances near the extent of the mouse’s reach. After an initial hold period, the animals were instructed to reach to a specific spout by the pitch of a cue tone, which occurred simultaneously with water delivery at that spout. The probability of each spout being cued was 50% on each trial, so the animal had no information as to where they should reach until the cue was played. After the cue, they performed rapid reaching and grasping movements to retrieve the water reward from the spout (Fig. 1B [↗](#)). The mice produced reach trajectories with substantial variability around the basic reach shapes (Fig. 1C [↗](#)), likely resulting from small differences in initial posture causing execution errors and corrections. Three design choices increased this movement complexity: (1) the water spouts were located near the maximal range of motion of each animal, (2) the water spouts were designed so that water would cling to them, and (3) we required good contact between the paw and the spout using touch sensors. As a result, the mice often displayed repeated grasp attempts to contact the spout and retrieve the entire water droplet (Fig 1G [↗](#)). This

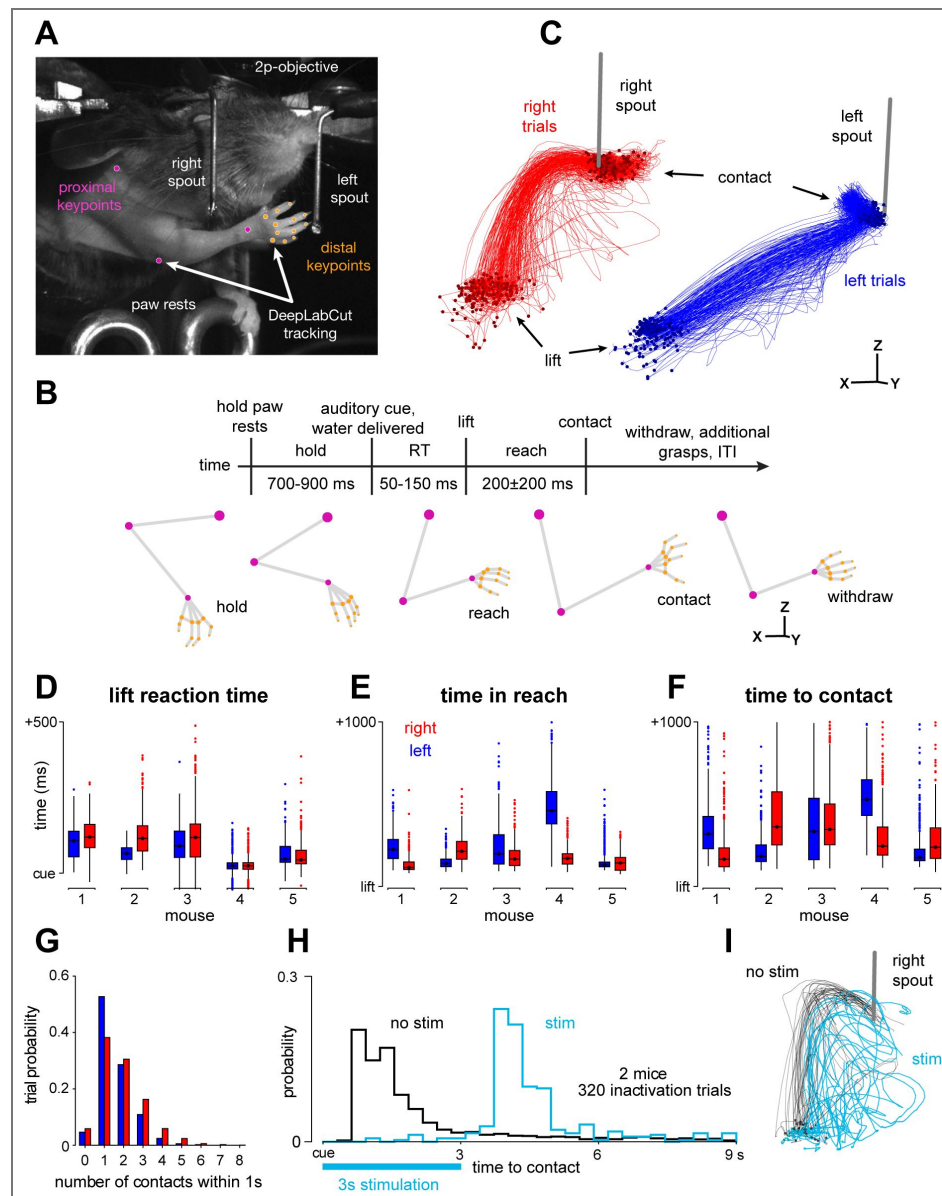


Figure 1. Mice make complex reaching and grasping movements to retrieve water rewards during head-fixation.

(A) Task setup: a head-fixed animal mid-reach with representative markerless tracking. (B) Task timeline, with representative 3D triangulated forelimb "skeletons". (C) Distal digit centroid trajectories from 100 ms before lift to 100 ms after contact. Stereo viewpoint corresponds to data image in A, left and right trial traces offset for visualization. One example session. (D) Lift reaction time distributions, left and right trials depicted separately. (E) Time from lift to paw centroid arriving within 5 mm of the spout target tip. (F) Time between lift and spout contact. Box plots in D-F show the quartiles (box and central line), whiskers extend 1.5 times the interquartile range past the quartile boundary, and more distant points (outliers) are plotted individually. (G) Number of contact events within 1 second of the water cue per trial. Data were pooled across sessions for D-G. (H) Time between cue and spout contact during optogenetic experiments. (I) Representative digit centroid trajectories for a subset of right trials during optogenetic experiments, from -100 ms before lift to 1000 ms after lift. Data from one example session.

variability across trials, spout targets, and mice was also seen in trial events like lift reaction time, reach duration, time-to-first-contact, and the number of contact events that occurred within 1 second of the cue on each trial (Fig. 1D-G).

In similar tasks, optogenetic inactivation of mouse M1 can halt the ongoing execution of reaching and grasping movements (Galiñanes et al. 2018; Guo et al. 2015). However, given that many of our mice performed the task for upward of 8 weeks and evidence that motor cortex disengages after extensive practice with lever pressing (Hwang et al. 2019, 2021; Kawai et al. 2015), it was not clear whether motor cortex would remain necessary in our task. To determine this, we performed optogenetic inactivations of the forelimb portion of M1 (M1-fl) (Muñoz-Castañeda et al. 2021) by activating inhibitory neurons in two VGAT-ChR2 mice. These inactivations blocked the execution of the reach to grasp sequence, preventing the animal from making contact with the spout during the 3-second laser stimulation period (Fig. 1F; 86.5% control trials with contact within 3 seconds of cue, 5.1% inactivation trials with contact, $P < 10^{-191}$, Mann-Whitney U test, 2 mice, 495 stimulation trials). Interestingly, inactivation at the time of cue often did not prevent reach initiation (mouse 1: 54.7%, mouse 2: 34.2% of inactivation trials with lift within 3 seconds; 93.5%, 86.2% control trials). Yet the movement stalled once the paw and digits extended towards the spout, producing uncoordinated and unsuccessful reaching trajectories (Fig. 1I, two representative datasets). Taken together, these results support the involvement of M1-fl in the water-reaching task and suggest that the strength of inactivation effects may depend on specific task details like training time or target configuration (Galiñanes et al. 2018).

Stereotyped aperture kinematics across idiosyncratic reach-to-grasp movements

Rodent forelimb movements have been described qualitatively as a set of sequential stereotyped postures when reaching to grasp food or water rewards (Whishaw and Pellis 1990) (Fig. 1B). We identified these features in our data using our high precision quantification of the mouse's proximal and distal forelimb during reaching. First, we defined a simple geometric measure of paw aperture that captured the strongest features of paw opening and closing, including the splay of the digits and flexion of the metacarpophalangeal (MCP) and proximal interphalangeal joints (PIP) (Fig. 2A). The first aperture minimum corresponded tightly with the peak of vertical paw velocity after lifting, with the paw reaching maximum vertical velocity shortly before the paw and digits became maximally flexed (Fig. 2B) ($r=0.6-0.95$ across mice, Pearson correlation; maximum velocity preceded minimum aperture by 27.2 ± 10.5 ms across mice). When aligning single-trial aperture traces to the peak vertical velocity (max Z velocity), we observed clear, stereotyped structure in how paw aperture evolved during the movements, across both spout targets for all five mice (Fig. 2E). This stereotyped shape was surprising given the heterogeneity in reach trajectories in this task and the strong differences in reaching strategies adopted for each target.

We termed the first aperture minimum after the lift to be the time of “collect,” following (Whishaw et al. 2010), and termed the next maximum after collect “extend” (Fig. 2B). These events structured the elaboration of the reach-to-grasp movement over time: after lifting, the paw closed to its minimum aperture (collect) before the digits were maximally opened (extend) as the paw was moved towards the spout. Extend occurred more heterogeneously in time after collect (Fig. 2C), perhaps reflecting single-trial variability in initial posture. These grasp-related events corresponded strongly with the reach's progress as well. Regardless of the initial paw location on the paw rest, lifting the arm brought the flexing paw to a consistent location at the moment of collect, from which the mouse could prepare an accurate extension movement towards the spout (Fig. 2D). These observations serve as quantitative confirmation of previous qualitative details obtained from high speed video and confirm rich structure in the mouse reach-to-grasp movement.

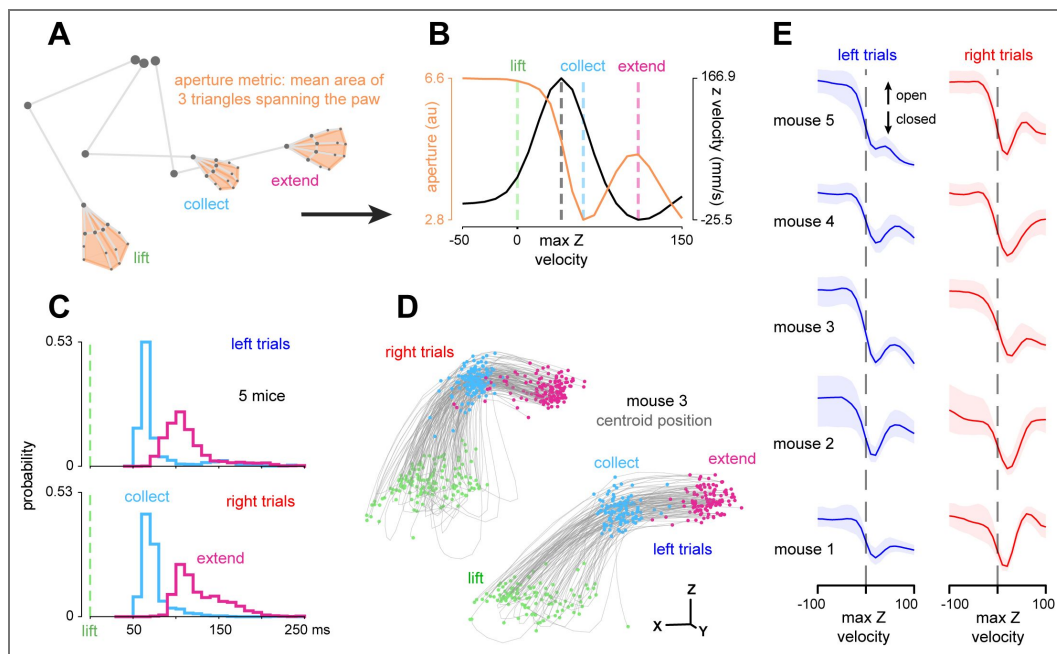


Figure 2. Paw aperture reveals stereotyped kinematics across idiosyncratic single trial reach-to-grasp movements.

(A) Paw aperture was computed from the paw markers using a simple geometric measure. (B) Aperture time series over-laid on the Z velocity of the paw centroid. The collect and extend events are defined by the aperture minimum and its subsequent local maximum. Representative trial. (C) Histograms of collect and extend event times for all trials, left and right trials separated. (D) Lift, collect and extend event times depicted on digit centroid trajectories for one example mouse. Left and right trials offset in X and Z as in Figure 1C. (E) Median over trials of aperture time series aligned to the moment of maximum centroid Z velocity.

Proximal and distal joints are coordinated during mouse reach-to-grasp

To obtain a high fidelity representation of the reach-to-grasp kinematics, we constructed a 3D “skeletal” representation of the forelimb by video-tracking keypoints at 15 putative joint centers along the proximal and distal axis (Methods). We then used axis-angle geometries to approximate 24 joint angles (7 proximal, 17 distal), including a number of intrapaw angles that captured deformations of the paw itself (Fig. 3A). These joint angles displayed rich temporal heterogeneity when reaching and grasping to both targets (Fig. 3B). Reaching evoked particularly large rotation of the shoulder, likely because the mice reached from a quadrupedal position to targets on either side of the snout. Joint angles were strongly correlated across the entire limb, consistent with the expectation that the forelimb is a system of joints that are controlled cooperatively and, in some cases, are mechanically linked (Fig. 3C). The strongest correlations occurred for angles describing the same joint (e.g., the 3 angles describing the shoulder) and for the flexion of MCP and PIP joints. The block-like structure of the joint angle correlation matrices indicated that control of the mouse distal limb is tightly linked to control of the proximal limb, consistent with previous observations of the strong stereotypy in mouse forelimb movements across behaviors (Naghizadeh et al. 2020; Whishaw 1996; Whishaw et al. 2010; Whishaw and Pellis 1990).

Despite the strong coupling between the proximal and distal joint angles, rich variation remained in the action of different joints over time. The presence of strong correlations across joints suggested that the kinematics may be well described by a smaller number of independent degrees of freedom than the total number of recorded angles. To assess the number of linearly independent (uncorrelated) degrees of freedom amongst the 24 joint angles and velocities, we used double-cross-validated PCA (Yu et al. 2009; Methods; Fig. 3D), finding intermediate dimensionalities of 7 (median for joint angles) and 10 (velocities; Fig. 3E). This is consistent with the idea that joint angles across the limb are coordinated instead of controlled independently, and that this coordination is flexible enough over time to enable accurately performing reaching and grasping to different targets.

Layer 2/3 cells in M1-fl and S1-fl are strongly modulated and heterogeneously tuned

Having characterized the structure of the mouse forelimb reaching and grasping movements, we next sought to determine how the kinematic details of these movements were distributed across the sensorimotor cortex. To do so, we recorded neural activity from neurons in layer 2/3 of M1-fl extending slightly into the immediately adjacent secondary motor cortex, and the forelimb region of S1 (S1-fl) using two-photon calcium imaging of GCaMP6f expressing neurons in layer 2/3 (185–230 μm deep, imaged at 31 Hz, cells extracted with Suite2p (Pachitariu et al. 2017)). A slight majority of recorded neurons in both areas (51.8% in M1-fl and 52.2% in S1-fl, $p < 0.05$ with Bonferroni correction for six tests) were statistically modulated to at least one event during the task: cue, paw lift, and/or first spout contact (Fig. 4B; see Methods, “ZETA”). Cells showed strong modulation locked to different task events, such as the cue onset (51.7% / 52.7% in M1-fl / S1-fl, left or right p -value < 0.05 corrected for dual tests), paw lifting from the rest (51.6% / 54.8% in M1-fl / S1-fl), and time of first contact with the spout (45.2% / 46.4% in M1-fl / S1-fl).

Cells were also modulated by the time taken to complete the reach: many cells exhibited different firing rates on trials with movement duration briefer versus longer than the median (Fig. 4C–D, see ROIs 35 and 116). Finally, many cells showed temporal heterogeneity in their activity beyond that of simple transient bumps. This included cells (such as ROI 54) that were strongly modulated by movement onset and remained active late into the trial. This activity may reflect the details of movement as the animals made repeated grasping attempts with a fully extended arm to retrieve the entire water reward from the spout.

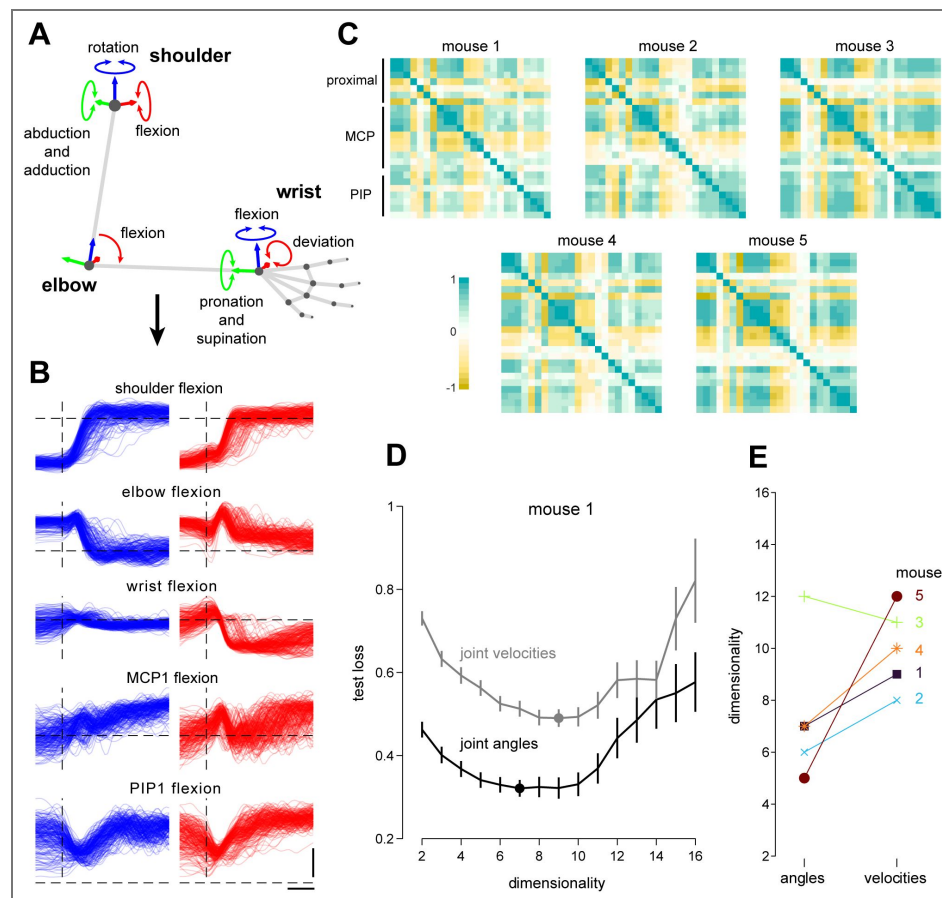


Figure 3. Mouse reaching and grasping movements involve coordination across proximal and distal joints.

(A) Schematic of Euler angle inverse kinematics procedure, depicting 7 joint angles over the proximal joints; 17 distal angles not shown. (B) Time series of 5 representative joints, shown from -100 ms to +400 ms around lift onset (vertical dashed line). Left (blue) and right (red) trials shown separately. Horizontal scale bar is 100 ms. Vertical scale bar is 30 degrees. Dashed horizontal lines are 0 degrees. (C) Correlation matrices of joint angles over all recorded time points. Rows are ordered descending from proximal (shoulder) to distal (metacarpal phalanges, MCP, proximal interphalanges, PIP). (D) Example loss curve of the cross-validated principal component analysis performed on joint angle and joint velocity time series. (E) Estimated dimensionalities of joint angle and velocity kinematics across mice.

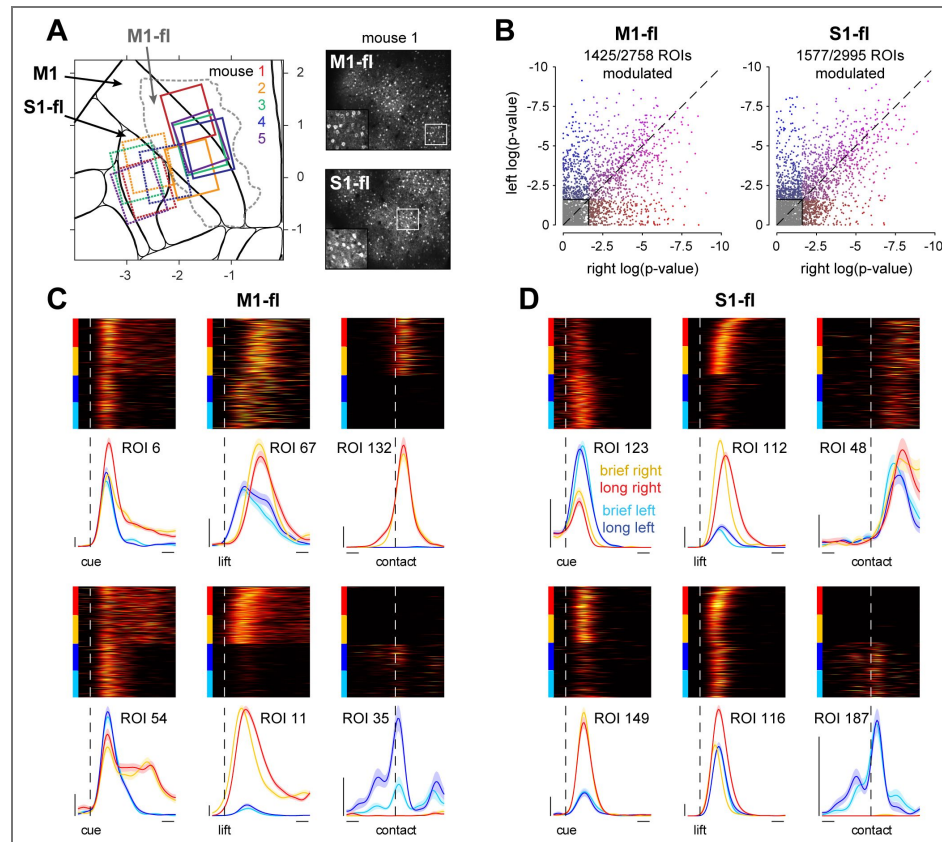


Figure 4. M1-fl and S1-fl cells are heterogeneously tuned to specific movement features.

(A) 2-photon imaging fields of view for all mice aligned to the Allen Atlas common coordinate framework (CCF). M1 corresponds to MOp in the CCF, and S1-fl corresponds to SSp-ul. The dashed gray line indicates the outline of the forelimb primary motor cortex (MOp-ul, referred to here as M1-fl) as described in Muñoz-Castañeda et al 2021 [DOI](#). See Figure 4S1 [DOI](#) and Methods for field of view alignment details. Right, example max-projection images for M1-fl and S1-fl for mouse 1 with magnified insets. (B) Modulation metric (see Methods) of cue-locked data for individual cells in M1-fl and S1-fl. Values displayed are the logarithm of the computed p-value, where each point represents a cell. All cells with p-values < 0.025 are shown colored: left-only modulated cells are blue, right-only modulated cells are red, and cells modulated to both conditions are purple. (C) Peri-event histograms and rasters of single trial activity from individual cells in M1-fl. Each column displays two cells locked to each trial event: cue, lift and first-spout-contact. Trials grouped within each condition by time-in-reach. Vertical scale bar: 10 events per second (arbitrary units). Horizontal scale bar: 100 ms. (D) As in C but for S1-fl.

Proximal and distal joint kinematics can be decoded from M1-fl and S1-fl population activity

To assess how movement related signals were distributed across M1-fl and S1-fl, we decoded the time series of each joint angle from neural population activity in each area. A simple linear regression model related the single-trial joint angles at all time points to single-trial neural activity at the corresponding moments. The model was fit with ridge regression, the ridge penalty was determined via a heuristic (Karabatsos 2018 [DOI](#)), and performance was measured on held-out trials (80/20 train/test split, 50 folds). We found that every measured proximal and distal joint angle could be decoded from M1-fl activity at above-chance levels during reaching and grasping (Fig. 5A-B [DOI](#)). Interestingly, S1-fl decoding exhibited similar decoding performance as M1-fl across all mice (Fig. 5E-F [DOI](#)). Including additional causal (−100 ms preceding) and/or acausal (−100 ms preceding to 100 ms following) lags improved decoding performance modestly and similarly for both areas (Fig. 5S3E-F [DOI](#)).

Across both areas we found that proximal joints were better decoded than distal joints (Fig. 5D,H [DOI](#)). A number of reasons for this difference are possible: this may reflect the time scale of 2p-recorded neural activity, overall smaller total variance of the most distal joint angles (see Fig. 5S2E-F [DOI](#)), or measurement noise present in the markerless tracking for small features like the fingertips. Nevertheless, we were able to explain substantial variance in the most distal intrapaw joints (see joints MCPf, MCPa, MCPo, PIPs and PIPf).

Additionally, we reconstructed the skeletal representation of the forelimb from the decoded joint angles and found that, as intended, the reconstructed postures had strong qualitative resemblance to the true postures, even of “minor” angles like cylindrical paw deformation or digit splay (Fig. 5C,G [DOI](#)). This suggests that movement-related signals that could be used to actuate forelimb joints are present in both M1-fl and S1-fl populations.

Finally, we tested whether the ability to decode these many joint angles was a direct consequence of inter-joint correlations, and might not be indicative of the presence of “real” information about some of these joints. To do so, we fit partial correlation models that removed correlations between proximal and distal joints, or removed correlations of the joint angles with a high-level parameter – the overall distance of the paw centroid to the spout. Despite substantially lowering the behavioral variance, in each case the residuals could still be decoded from neural activity (Fig. 5S2A-D [DOI](#)). Similar decoding performance for M1-fl and S1-fl was obtained from models fit to decode single-trial residuals separately for left and right trials (Fig. 5S3A-D [DOI](#)), indicating that trial-to-trial variations on each basic movement were decodable from these populations.

Overall, these results reveal that neural activity in M1-fl and S1-fl is closely related to the kinematic details of reach-to-grasp movements. The ability to decode substantial variance in proximal and distal joints suggests that this relationship extends to the entire forelimb, and the similar performance obtained from each area suggests that this information is similarly distributed across M1-fl and S1-fl.

M1-fl reflects target-specific information earlier and more persistently than S1-fl

Given the similar levels of kinematic information in M1-fl and S1-fl, we sought to determine if these areas differed in their encoding of a higher-level aspect of movement, target location. To assess how target-related signals were organized in each area, we analyzed how well target identity could be decoded from neural activity over time. For each session, a linear classifier was trained to predict whether the neural activity from a particular window of time came from a left-target or right-target trial. This decoder was then applied to held-out trials from the same time window. This analysis revealed that M1-fl activity reflected the target location slightly more faithfully than did S1-fl, as shown by its higher overall performance (93.2% M1-fl vs. 89.1% S1-fl median over cross-validation folds and time bins; $p = 4.1 \times 10^{-4}$, Wilcoxon rank sum test). To examine single trials, we then projected each trial's population activity onto the identified

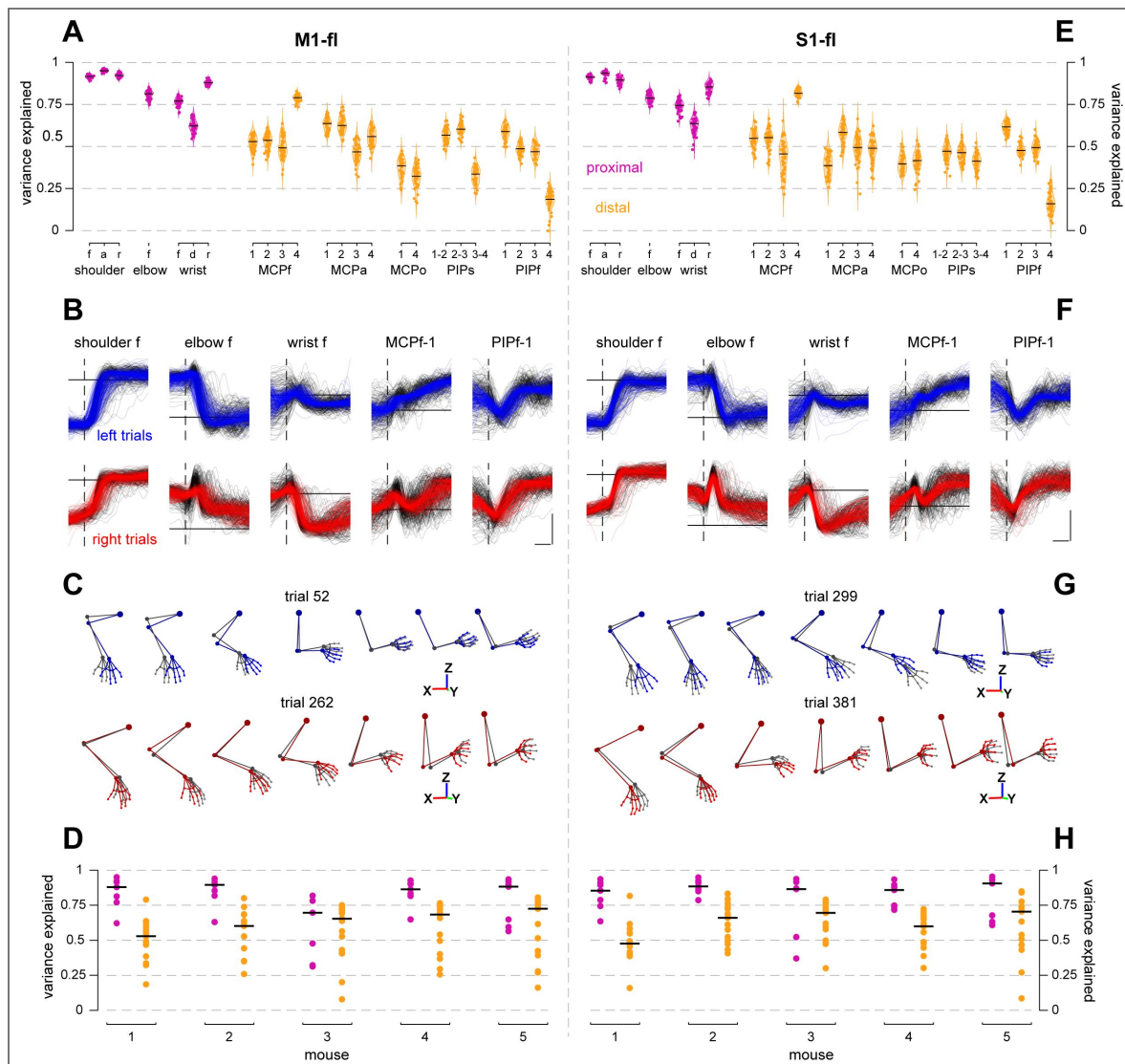


Figure 5. Kinematic details of proximal and distal joints during reaching and grasping can be decoded from M1-fl and S1-fl population activity.

(A) Performance of decoding models predicting joint angle kinematics from RADICaL-inferred M1-fl rates. Violins depict the distribution of variance explained over cross-validation folds for each individual joint angle. (f: flexion, a: abduction/adduction, r: rotation, d: deviation, o: opposition, s: splay, MCP: metacarpal-phalanges, PIP: proximal-interphalanges; digits indicated by number). (B) Reconstructed joint angle time series for 5 representative joint angles. Black traces are original and colored are reconstructed; left trials are blue and right trials are red. Vertical dashed line indicates lift time, horizontal line indicates 0 degrees. Vertical scale bar is 30 degrees, horizontal scale bar is 100 ms. (C) Forelimb skeletal postures reconstructed from decoded joint angles for representative left and right trials (trials having median variance explained). 7 representative time points from lift time to grab time are shown. Gray postures are original data, colored postures are reconstructed as in B. Decoded joint angles in each time step were applied to the neutralized posture of the original data, preserving the original lengths of inter-marker links. (D) Performance of decoding models for all 5 mice. Each point is the median of the distribution of variance explained over cross validation folds, with proximal (magenta) and distal (goldenrod) angles considered separately. The black lines indicate the median variance explained over all proximal or distal joints for an individual mouse. (E-H) As in A-D but for S1-fl data.

decoding axis (the “target-specific dimension”). In this projection, M1-fl activity predicted the target sooner after the cue than did S1-fl activity (59.9 ± 34.7 ms earlier, $p < 0.05$ for all mice, Wilcoxon rank sum test) and displayed a stronger, more consistent, and longer-lasting prediction throughout the trial (Fig. 6A,B).

Target decoding performance could result from truly higher-level signals that code abstractly for target location, or alternatively could be supported by strong encoding of kinematic variables that differed between targets. To disambiguate these possibilities, we refit the linear classifier to neural data after regressing off variance related to the joint angle kinematics. The strength and exact time course of the resulting target decoding varied somewhat across animals, but the earliest portion of target decoding performance persisted in all animals after the removal of kinematics and performance remained stronger for M1-fl than S1-fl (Fig. 6S1B). We thus conclude that higher-level signals are present in both areas, but differ in their exact timing and strength. However, we note that other possible signals, such as postural changes, could not be controlled for here.

To determine whether the target-specific dimension was conserved or whether it changed over the course of a movement, we then tested how well a decoder trained during one time window generalized to held-out data from other time windows. M1-fl decoders were more stable over the trial, and performed better when generalizing to far-away time points than did S1-fl decoders (Fig. 6C). These results suggest that the encoding of target-specific information is more consistent over time in M1-fl than S1-fl.

Finally, we asked how strongly each area encoded a different high-level parameter, resting versus reaching. To do so, we trained a decoder to predict whether neural activity was from a time point pre- or post-lift, then computed a movement onset time from when the decoder switched from pre- to post- (Methods; Fig. 6D). Unlike our results for predicting target identity, S1-fl activity more strongly predicted lift time than did M1-fl ($p = 0.0079$, Wilcoxon rank sum).

Discussion

The extent to which activity in sensorimotor cortex reflects the details of movement has remained unclear in the mouse. Here we show that in a challenging reach-to-grasp task evoking variable movements, both M1-fl and S1-fl relate tightly to all the joint angles we could track throughout the arm, paw and digits. Using high-fidelity markerless tracking, we replicated prior, coarser results that the reach-to-grasp kinematics have stereotyped structure and involve strong coordination across proximal and distal joints. Using two-photon calcium imaging, we revealed that single neuron activity in M1-fl and S1-fl is strongly modulated and tuned to fine movement features. Decoding models then demonstrated that M1-fl and S1-fl both possess rich representations of low-level joint angle information at similar levels of fidelity. Despite these similarities, target decoders revealed that high-level movement information was stronger and more persistent at the population level in M1-fl than S1-fl. Activity in S1-fl was instead more transient and more closely linked to lift onset than in M1-fl.

These results expand our understanding of the rodent sensorimotor system and highlight similarities to nonhuman primates. We show here evidence in mice of detailed joint angle kinematic signals from the full forelimb in M1 and S1, as has been shown in macaque cortex during tasks involving reaching and grasping objects (Vargas-Irwin et al. 2010; Saleh et al. 2010, 2012; Goodman et al. 2019; Okorokova et al. 2020). Additionally, the earlier onset of movement-related activity in M1-fl compared to S1-fl is similar to macaque M1 and S1 (Tanji and Evarts 1976). Taken together these results suggest that the mouse can be employed to address questions traditionally explored in primates about how cortical activity encodes detailed movement commands.

The strength of movement-related activity in these areas is surprising given the results of loss-of-function experiments in rodents, which have suggested that sensorimotor cortex can become disengaged from forelimb control with extensive training (Hwang et al. 2019, 2021; Kawai et al. 2015), and that subcortical structures like the medulla or basal ganglia can independently

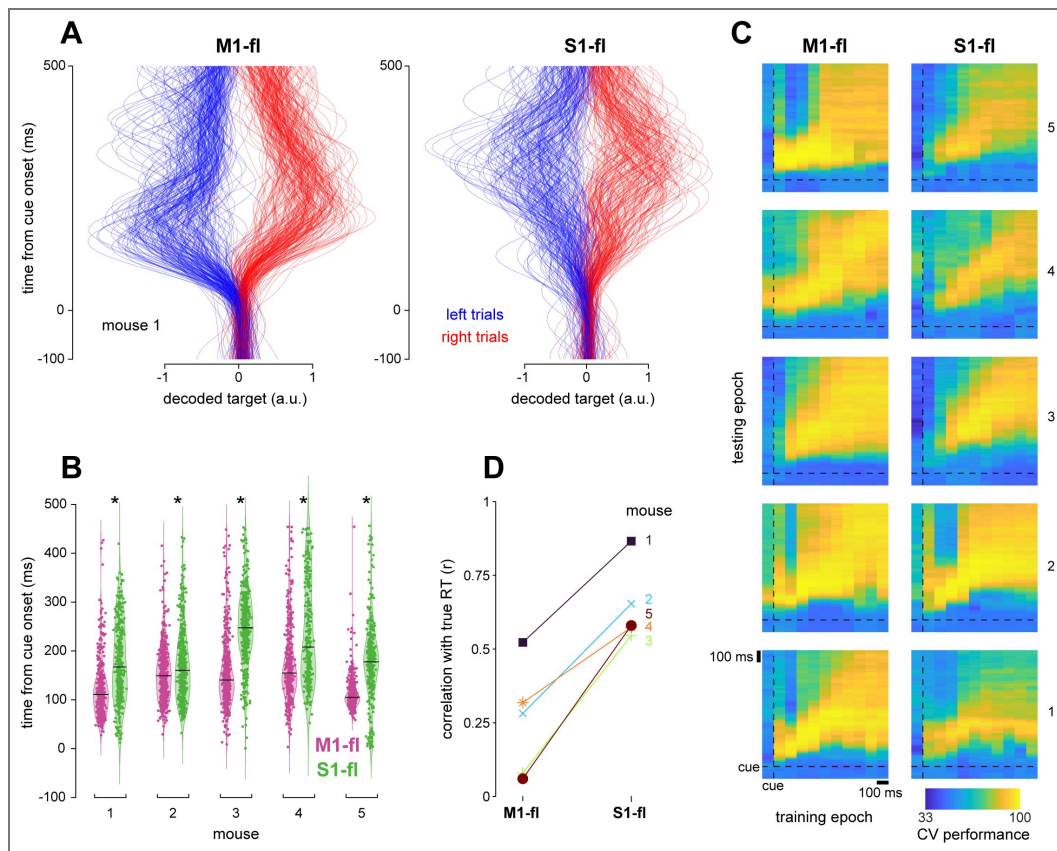


Figure 6. M1-fl reflects target-specific information earlier and more persistently than S1-fl.

(A) Projections of population activity from held-out single trials onto the target decoding dimension for M1-fl and S1-fl. Right trials are red and left trials are blue. Traces were normalized by the 90th percentile value across trials for left and right trials separately. (B) Distribution of threshold crossing times for projections shown in A. Black lines indicate medians. Star indicates significantly different medians at $p < 0.05$, Wilcoxon ranksum. (C) Performance of time-generalized target decoding analysis. Each heat map depicts the performance over all model training windows (columns) and test time points (rows). Time in trial runs vertically as in A. Both the training and testing epochs were -100 (leftmost column training, bottommost row test) to $+1000$ ms (rightmost column training, top row test) relative to cue. Training epochs were 100 ms non-overlapping windows and testing epochs were 10 ms bins. (D) Performance of movement onset prediction for M1-fl and S1-fl. Correlation coefficient r was computed between the true and predicted RTs across held-out trials.

control coordinated forelimb movements (Esposito et al. 2014 [↗](#); Ruder et al. 2021 [↗](#); Yang et al. 2023 [↗](#)). However, several other previous results have indirectly suggested that M1 and S1 may be involved in the details of forelimb movement. Performance suffers with inactivation or lesioning of M1 and S1 in skilled, complex manual behaviors (Guo et al. 2015 [↗](#); Mizes et al. 2024 [↗](#); Whishaw et al. 1991 [↗](#)) or idiosyncratic use of digits to accomplish non-dexterous tasks (Kawai 2014 [↗](#)). The sparing of non-dexterous tasks with these lesions may also reflect redundancy in control as opposed to irrelevance of M1 and S1. Nevertheless, our finding of low-level kinematic information in sensorimotor cortex supports a role for cortex beyond simply providing redundant high-level commands to these subcortical areas.

There are a number of possible roles for this movement-related cortical activity. One is that motor cortex may dominate control when the behavior is complex enough in some way – like when a cue is needed to identify the target (Mizes et al. 2024 [↗](#)) or when sensory feedback is necessary to coordinate dexterous grasps in the presence of errors (Guo et al. 2015 [↗](#); Perich et al. 2024 [↗](#); Sauerbrei et al. 2020 [↗](#)). Despite the overtraining of our behavior, these task features might preclude use of the presumably less-flexible subcortical circuits alone. Alternatively, subcortical areas like the medulla or cerebellum may initiate the movement (Dacre et al. 2021 [↗](#); Esposito et al. 2014 [↗](#); Inagaki et al. 2022 [↗](#); Ruder et al. 2021 [↗](#)) while the cortex elaborates the details of otherwise-generic commands (Kaufman et al. 2016 [↗](#)). This might permit the previously-observed disengagement of cortex with learning of simple behaviors, could explain our observations of partially executed reach-to-grasp movements during M1-fl inactivation, and would explain the weak representation of lift-onset in our M1-fl data. A third alternative is that medulla alone executes low-level control, but low-level information is maintained in sensorimotor cortex for some other function, such as to enable learning or error correction (Kawai et al. 2015 [↗](#); Makino et al. 2016 [↗](#); Wolff et al. 2022 [↗](#); Kirk et al. 2024 [↗](#); Koh et al. 2025 [↗](#)). These hypotheses are not mutually exclusive.

Our observation of detailed kinematic signals in the mouse forelimb somatosensory cortex raises the possibility of a central role for S1-fl in dexterous movements, and may implicate a more broadly distributed circuit in forelimb control than previously considered. Indeed, the magnitudes of representation in M1-fl and S1-fl were nearly identical. This may simply be a consequence of widely distributed representations of movement across mouse cortex (Musall et al. 2019 [↗](#); Steinmetz et al. 2019 [↗](#); Stringer et al. 2019 [↗](#)), including forelimb somatosensory areas, or may be a consequence of the close physical proximity of M1-fl and S1-fl hindering development of functionally distinct representations (Tennant et al. 2011 [↗](#)). However, the strength of detailed kinematic information in S1-fl supports a role beyond simply receiving a broad corollary discharge signal (Fromm and Evarts 1982b [↗](#)).

One possibility is that proprioceptive signals are strong enough to carry these movement details into S1-fl from subcortical structures, as is true in primate forelimb S1 (Umeda et al. 2019 [↗](#)). Our lift time decoding results are consistent with this view and align with recent observations characterizing mouse proprioceptive forelimb cortex (Alonso et al. 2023 [↗](#)), although an alternative explanation may be simply that M1-fl activates earlier than S1-fl during reaching (Kargo and Nitz 2004 [↗](#); Miri et al. 2017 [↗](#); Veuthey et al. 2020 [↗](#)). Further, our task evoked repetitive grasping attempts, which may engage S1-fl strongly to inform motor areas of the sensory consequences of movement (Perich et al. 2024 [↗](#)). Extensive circuit tracing has shown that forelimb M1 and forelimb S1 are highly interconnected in the mouse (Muñoz-Castañeda et al. 2021 [↗](#); Yamawaki et al. 2021 [↗](#)), suggesting that detailed movement signals in S1-fl may reflect not only sensory feedback but also efference copies of motor commands generated by M1-fl. Rodent S1 is known to influence interneuron populations in the spinal cord through direct and indirect projections that predominantly target the dorsal horn (Ueno et al. 2018 [↗](#)), and thus these signals may also reflect S1-fl's important role in modulating reflex circuits to coordinate sensory feedback with movement generation (Moreno-López et al. 2016 [↗](#); Moreno-Lopez et al. 2021 [↗](#); Seki et al. 2003 [↗](#)).

Several limitations of the present work should be noted. Pose estimation was performed with state-of-the-art neural networks (Mathis et al. 2018 [DOI](#)) and extensive hand-labeling of keypoints together with quality assurance metrics, but behavior tracking is inevitably imperfect. In particular, a noise floor remains that particularly impacts small bones such as the distal phalanges, and markers estimated at the skin surface cannot perfectly reflect underlying joint and muscle kinematics. Additionally, calcium imaging has limited temporal resolution and is a nonlinear transformation of spiking activity (Stringer and Pachitariu 2019 [DOI](#); Wei et al. 2020 [DOI](#)). Although we used leading-edge deconvolution methods (Friedrich et al. 2017 [DOI](#)) and cutting-edge neural network approaches for sub-frame event estimation (Zhu et al. 2022 [DOI](#)), these limitations cannot be entirely overcome. These slow temporal dynamics limited our ability to disambiguate whether joint angles or joint velocities are more linearly encoded in these areas. The use of calcium imaging further prevents strong conclusions about whether activity reflects future limb states or sensory consequences. Confirming this limitation, inclusion of lagged data in the decoding models, whether causal or acausal, resulted in similar performance changes in both areas. Moreover, as is widely known (Todorov 2000 [DOI](#)), the exact role of these kinematically-related signals is challenging to determine from correlative measures alone; thus, determining whether these signals are used for direct movement control or indirectly reflect control performed elsewhere is a topic for future work. Nevertheless, our ability to decode substantial variance from small joint angles and fast changes within the paw supports the claim that movement signals are represented in M1-fl and S1-fl activity. Finally, all recordings included here were taken from layer 2/3 of both M1-fl and S1-fl. Activity may differ in layer 5, where the primary subcortical outputs originate.

Overall our results suggest that mouse sensorimotor cortex is intimately involved in the control of dexterous forelimb movements, and support use of the mouse as a first-class model system for studying the cellular and circuit basis of motor control. The presence of detailed kinematic signals in the sensorimotor cortex supports a model of mouse sensorimotor cortex in which M1-fl and S1-fl play a role in shaping the fine details of reaching and grasping movements. Further work will be required to map the roles of other sensorimotor cortical areas, understand their interactions, and differentiate between ballistic reaching and feedback-driven grasping and withdrawal.

Methods

Animal subjects

All procedures were approved by the University of Chicago Institutional Animal Care and Use Committee. A portion of the data presented here were analyzed in a previous study (mice 1 and 2; (Zhu et al. 2022 [DOI](#))). For two-photon imaging, five male male Ai148D transgenic mice (TIT2L-GC6f-ICL-tTA2, stock 030328; Jackson Laboratory) aged 8-12 weeks were used. Four of the mice carried the GCaMP6f transgene heterozygously and one homozygously (mouse 3). For optogenetic experiments, 2 VGAT-ChR2 (stock 014548; Jackson Laboratory) transgenic mice aged 8-12 weeks were used. Each animal underwent a single surgery. Mice were then individually housed in a reverse 12-h light/dark cycle, with an ambient temperature of 71.5 °F and a humidity of 58%. Experiments were conducted in the afternoon, during the animal's dark cycle.

Surgical procedures

Mice were injected subcutaneously with dexamethasone (8 mg kg^{-1}) 24h and 1h before surgery to reduce inflammation. Mice were anesthetized with 2–3% inhaled isoflurane gas and then injected intraperitoneally with a ketamine–medetomidine solution (60 mg kg^{-1} ketamine and 0.25 mg kg^{-1} medetomidine). Mice were given supplemental low level isoflurane (0–1%) if they showed any signs that the depth of anesthesia was insufficient or began emerging from anesthesia towards the end of the surgery. Animals received a subcutaneous injection of meloxicam (2 mg kg^{-1}) at the beginning of the surgery and once daily for 2 days post-operation. The scalp was shaved, cleaned and resected, then the skull was cleaned and the wound margins glued to the skull with tissue glue (VetBond, 3M). A circular craniotomy was then made with a 3-mm or 4-mm biopsy punch centered

over the left MOP / SSp-ul border, as defined by the Allen Common Coordinate Framework v3 (Allen CCF). The coordinates for the center of M1-fl were taken to be 0.4 mm anterior and 1.5 mm lateral of bregma. The craniotomy was cleaned with SurgiFoam (Ethicon) soaked in phosphate-buffered solution (PBS). If the dura remained intact, a durotomy was performed.

Virus (AAV9-CaMKII-Cre, diluted to 1×10^{13} particles per mL in PBS, Addgene stock 105558-AAV9) was then pressure injected (Nanoject III, Drummond Scientific) at 2-6 sites near the target site, with 140 nL injected at each of two depths per site (250 and 500 μ m below the pia) over 5 min each. The injection pipette was kept in place for another 3-6 minutes to ensure viral dispersion before being removed slowly over 2-3 minutes. In mice numbers 3-5 an additional virus injection (AAVretro-tdTomato, diluted to 1×10^{13} particles per mL in PBS, Addgene stock 59462-AAVrg) of 60-100 nL was made targeting the bulk of forelimb somatosensory cortex at -0.73 ± 0.08 mm posterior and $\sim 2.69 \pm 0.15$ mm lateral of bregma.

This injection retrogradely labeled cell bodies that sent projections to S1-fl. In this paper the labeling was used solely for stabilizing the imaging plane (see below) in both areas. In mice 1 and 2, a small craniotomy was made using a dental drill over the right M1-fl at 0.4 mm anterior and 1.6 mm lateral of bregma. 140 nL of AAVretro-tdTomato (1×10^{13} particles per mL, Addgene) was then injected at 300 μ m below the pia. This injection labeled cells in the left M1-fl projecting to the contralateral M1-fl. As above, this labeling was used solely for stabilizing the imaging plane.

The injection craniotomy was sealed with a drop of Kwik-Cast (World Precision Instruments). For mice 1-3 and the two VGAT-ChR2 mice the cranial window craniotomy was sealed with a custom cylindrical glass plug (3 mm in diameter, 660 μ m in depth; Tower Optical) bonded (Norland Optical Adhesive 61, Norland) to a 4-mm number 1 round coverslip (Harvard Apparatus) that was glued to the skull surface with tissue glue. For mice 4 and 5 the craniotomy was sealed with a custom metal and glass cannula created from a small stainless steel ring (3.96 mm outer diameter, 3.45 mm inner diameter, 0.76 mm length, MicroGroup) bonded to a 4 mm number 1 round coverslip using cyanoacrylate glue (ZAP-A-GAP, Robart). All window implants were glued in place first with tissue glue (VetBond) and then with cyanoacrylate glue (Krazy Glue) mixed with dental acrylic powder (Ortho Jet; Lang Dental). In all mice two layers of MetaBond (Parkell) were applied, and then a custom laser-cut titanium head bar was affixed to the skull with black dental acrylic. Animals were awoken by administering atipamezole via intraperitoneal injection and allowed to recover at least 3 days before water restriction.

Behavioral task

As previously reported (Zhu et al. 2022 [DOI](#)), the behavioral task (Fig. 1 [DOI](#)) was a variant of the water-reaching task of (Galiñanes et al. 2018 [DOI](#)) that we term the ‘water grab’ task. This task was performed by water-restricted, head-fixed mice, with the forepaws beginning on paw rests (eyelet screws) and the hindpaws and body supported by a custom, 3D-printed, tapered, clear acrylic tube enclosure. After holding the paw rests for 700–900 ms, a tone was played by stereo speakers and a 2- to 3- μ L droplet of water appeared at one of two water spouts (22 gauge, 90-degree bent, 1-in blunt dispensing needles, McMaster) positioned on either side of the snout. The pitch of the tone indicated the location of the water, with a 4 kHz tone indicating left and a 7 kHz tone indicating right. The cue tone lasted 500 ms or until the mouse made contact with the correct water spout, whichever came first. The mouse could grab the water droplet and bring it to its mouth to drink any time after the tone began. Both the paw rests and spouts were wired with capacitive touch sensors (Teensy 3.2, PJRC). Good contact with the correct spout produced an intertrial interval of 3–6 s, while failure to make contact (or insufficiently strong contact) with the spout produced an intertrial interval of 20 s. Because the touch sensors required good contact from the paw, this setup encouraged complex contacts with the spouts. Control software was custom written in MATLAB R2018a using PsychToolbox 3.0.14. Touch event monitoring and task control were performed at 60 Hz using the Teensy.

Behavioral task training

The mice were trained to make all reaches with the right paw and to keep the left paw on the paw rest during reaching. For the first 1-3 days, mice were head-fixed and provided water directly to their mouths to acclimate them to head-fixation and reward collection. During this period the animals were encouraged to make consistent contact with both paw rests to trigger water delivery at their mouths. After acclimation and consistent trial initiation the spout was withdrawn to a position on the right side of their face intended to elicit forelimb reaching with the right arm. If animals were reluctant to reach, small droplets of water were placed on their right whisker pad with a blunt syringe to induce grooming and water collection. After animals reliably performed reaches toward the rightward spout location (>80-90% success), another spout was installed near the initial location (the “left” spout). Water was then solely delivered through the left spout and its position was shifted across the midline to the ventral contralateral whisker pad over the course of 7-10 days. Once the left spout was in a nearly-equidistant location from the midline as the right spout, the animal was cued to reach to both spouts in blocks of 10-50 trials. After reaching >90% in blocked trials, trials were cued randomly. This training procedure took 2-4 weeks, although the behavior continued to solidify for at least two more weeks. Data presented here were collected after 6–8 weeks’ experience with the task.

Optogenetic inhibition

Two VGAT-ChR2 mice were trained to perform the task. After 6-8 weeks of training, inactivations were performed using the epifluorescence path of our 2-photon microscope, centered over coordinates 1.5L 0.5A from bregma. The stimulation protocol was designed to approximately replicate the experiments of (Guo et al. 2015 [link](#)). Pulses were 17 mW peak power, 12.5 ms long and delivered at 40 Hz (50% duty cycle, 8.5 mW average power). The spot of light was approximately 1.3 mm in diameter at the focal plane. Stimulation was delivered beginning 150 ms before the cue and then pulsed continuously for 3 seconds after the cue. After an initial behavioral warm-up period of 50 control trials, stimulation was delivered with 25% probability on subsequent trials. To compensate for visible blue light due to leak around the light-sealing cone or through the brain, two bright blue LED strips were placed in front of the animal and illuminated constantly throughout the session.

High speed videography and stereo triangulation

As previously reported (Zhu et al. 2022 [link](#)), high-speed video data was recorded using a pair of cameras (BFS-U316S2M-CS, FLIR; varifocal lenses COZ2813CSIR2, Computar) mounted 150 mm from the right paw rest at 10° apart. Infrared illuminators enabled behavioral imaging while performing 2p imaging in a darkened microscope enclosure. Cameras were synchronized and recorded at 150 frames per second with real-time image cropping and JPEG compression and streamed to one HDF5 file per camera (areaDetector module of EPICS, CARS). DeepLabCut (DLC) (Mathis et al. 2018 [link](#)) was used to track 15 keypoints in each image (see next section). The MATLAB Stereo Camera Calibration toolbox was used for estimation of stereo camera parameters to triangulate paired keypoints into 3D.

Markerless tracking

3-5 days before recording, the right forelimb of each mouse was shaved and depilated using Nair while the animal was under light anesthesia (1-2% isoflurane). We did not observe any obvious impacts of this depilation on the behavior. DLC was then used to identify 15 keypoints associated with bony landmarks spanning the proximal and distal forelimb. These keypoints included: the tip (#1-4), proximal interphalangeal joints (#5-8) (PIP) and metacarpal phalangeal joints (#9-12) (MCP) of each digit, the center of the wrist (#13), distal edge of the elbow joint (#14), and proximal tip of the lesser tubercle of the humerus (#15). We developed a custom GUI in Python to improve the quality of manually labeled training frames. This GUI allowed simultaneous labeling in both images and visualized the epipolar lines of each keypoint in each image to improve stereo correspondence. We then employed an active learning approach to create a dataset of manually

labeled images. First, 20-80 frames paired between cameras were randomly selected from all 12 datasets. These images were manually labeled for all 15 markers and an initial DLC model was trained to identify keypoint locations for both cameras. We then performed 3 rounds of active learning, selecting 15-35 additional paired frames per dataset per round. We selected frames for inclusion in each round by identifying 3D postures in each video that strongly differed from those already contained in the training set (Feng et al. 2023 [DOI](#)). We assessed the similarity of candidate frames in a window 300-1000 ms around the water cue to the training set by computing the sum of Euclidean distances between vectorized 3D keypoint positions. We then sorted candidate frames by this distance and manually selected additional frames to label from the 10% of candidate frames with the greatest total distance. At the end of this process we labeled a total of 1455 paired images across 12 datasets (100-165 images per dataset) for a total of 2910 high quality labeled images.

Joint angle kinematics

To extract joint angle kinematics from the 3D triangulated keypoints, we first obtained vectors connecting adjacent keypoints in the unsmoothed data. Because we chose keypoint locations to closely align with putative joint centers across the forelimb, the angles between vectors connecting keypoints were taken to approximate the angles associated with the joint.

For the shoulder joint, we identified the “humerus” with the vector pointing from the shoulder keypoint to the elbow keypoint. For the elbow joint we identified the “radius/ulna” with the vector from elbow to wrist. The vector used to compute wrist angles pointed from wrist to the mean of the second and third MCP keypoints (virtual MC or vMC). For each digit, we computed the vector connecting the corresponding MCP to PIP keypoints, and finally the PIP to DIP keypoints. The vectors defined in this way were used to compute extrinsic Euler angles around each of the putative joint centers in a proximal-to-distal fashion. After each joint angle was computed, the corresponding vector and all of its children vectors were rotated so as to neutralize the rotation around that joint.

We chose a coordinate system appropriate for our task in mice. Starting with the shoulder joint, we computed the abduction/adduction of the humerus as the rotation around the external (world) Y axis (green vector in [Fig. 3A](#) [DOI](#)) necessary to bring the humerus into the ZY plane; abduction was chosen to be negative and adduction was positive. Next we computed the flexion/extension as the rotation around the X axis (red vector in [Fig. 3A](#) [DOI](#)) required to align the humerus with the Z axis (blue vector in [Fig. 3A](#) [DOI](#)); flexion was positive. Finally, we computed the shoulder rotation as the angle around the Z axis required to bring the radius/ulna into the YZ plane; internal rotation was positive. At the elbow we computed flexion/extension as the angle around the X axis required to make the radius/ulna parallel to the Y axis; flexion was positive.

We followed a similar procedure for the wrist as the shoulder; abduction/adduction was computed as the angle around the X axis required to bring the vMC into the XY plane; radial deviation or adduction was positive. We computed the wrist flexion/extension as the angle around the Z axis that brought the vMC into the YZ plane; flexion was positive. Finally, to compute the wrist rotation we first obtained the vector normal to the plane spanned by the second and third MCP vectors (nMCP). Wrist “rotation” (torsion of the radius around the ulna) was computed as the angle around the Y axis required to bring the nMCP into the YZ plane; pronation was positive. We then computed two intermetacarpal angles, quantifying how “opposed” MCP1 or MCP4 was to MCP2 or MCP3 respectively. This measure quantifies the conical deformation of the paw. This was done by computing the angle around the Y axis required to bring the MCP1 or MCP4 vectors into the YZ plane so that they were coplanar with MCP2 and MCP3. We calculated paw splay for the metacarpal 1-2, 2-3, and 3-4 vectors as the angle around the Z axis required to bring them into the XY plane. We then computed two angles around each MCP joint: abduction/adduction and flexion/extension. MCP abduction/adduction was computed as the angle around the X axis required to bring the proximal phalanx into the XY plane; abduction from the wrist midline was

positive. We computed the metacarpal flexion/extension as the angle around the Z axis required to bring the proximal phalanx vector into the YZ plane; flexion was positive. Finally, we computed the PIP flexion as the angle between the distal phalanx and the proximal phalanx for each digit.

Trial screening

To identify trials with problematic keypoint tracking, we used the extracted joint angle kinematics. We first binned joint angles for each trial at 10 ms in a window from -100 ms to 400 ms locked to lift onset and smoothed with a 15 ms Gaussian kernel. Left and right trials were considered separately. For each joint angle we found the 2.5 and 97.5 percentile values (pctile1 and pctile2) across all trials and time points. We then computed threshold values for each joint angle as $(\text{pctile1} + \text{pctile2}) / 2 \pm \text{abs}(\text{pctile1} - \text{pctile2})$. Across all trials, we identified individual joint angle values that exceeded the bounds of these two thresholds. Next, for each trial we computed the number of time bins for which any joint angle was outside the bounds. If a trial had more than 5 such time bins it was excluded from further analysis. Across all mice this procedure identified 4719/4993 trials with high quality tracking, which were further confirmed by visual inspection. After screening, no substantial tracking errors were detected even on single frames, and thus no further cleaning was performed.

Joint angle correlations and dimensionality

As in the screening procedure we first binned at 10 ms, locked joint angles from -100 ms to 400 ms relative to lift onset, and smoothed each angle within trials with a 15 ms Gaussian kernel. We then concatenated all trials across time. To exclude unusual joint angle configurations due to contact forces, we removed time points during which the paw made contact with the spout target. We then computed the correlation matrix shown in Figure 3 [using the *corrcoef* MATLAB function on this concatenated matrix](#). To compute the dimensionality of the joint angles and joint velocities, we performed double cross-validated principal component analysis (PCA) on the same concatenated matrix (Yu et al. 2009 [using the *corrcoef* MATLAB function on this concatenated matrix](#)). To perform this procedure, we first held out a subset of observations and performed PCA. Then, for each held-out observation, we held out one joint, used the values of the other joints together with the loadings to estimate the low-D coordinates on that observation via regression, and finally estimated the value of held-out joint in the held-out observations from the low-D coordinates and the loading for that joint. After cycling over all variables and 5 random observation partitions for each candidate dimensionality, we identified the optimal dimensionality as the one that minimized the reconstruction error of the held-out variable in the held-out observations. This procedure was performed separately for joint angles and joint velocities.

Wide-field imaging for area identification

To identify the forelimb primary somatosensory cortex, we performed widefield imaging while providing sensory input to the contralateral paw using a vibration motor (SparkFun, part ROB-08449) glued to a paw rest (Alonso et al. 2023 [using the *corrcoef* MATLAB function on this concatenated matrix](#)). This provided vibrotactile stimulation of the entire forelimb. The mouse was in the type of behavioral setup they were accustomed to except for the paw rest, and did not have to perform any task. Vibration was delivered for 250 ms once the mouse's paws had been on the paw rests for at least 2 seconds.

The widefield macroscope was identical to the setup of (Musall et al. 2019 [using the *corrcoef* MATLAB function on this concatenated matrix](#)), except that a pco.panda 4.2 sCMOS camera (Excelitas) was substituted and LED triggers were delivered by a Teensy 3.2. Imaging was performed at 30 frames/s, alternating blue and violet illumination. Violet frames were used for hemodynamic correction.

We used the locus of stimulation-evoked activity to align each cranial window to the Allen CCF. The location of strongest activation was aligned with the center of the widest part of the CCF area named SSp-ul (roughly 0 AP 2.6 ML relative to bregma), where proprioceptive and tactile responses most strongly overlap. We targeted our S1 imaging fields of view to this location as done in (Alonso et al. 2023 [using the *corrcoef* MATLAB function on this concatenated matrix](#)) and referred to the recorded region as forelimb S1 or S1-fl. With the window aligned to the atlas, we identified M1-fl as a compromise between this alignment and the coordinates of 0.4 AP and 1.6 ML relative to bregma. We targeted our M1-fl imaging field of views

to this region. In some cases, thick vasculature or weaker indicator expression prevented high-quality imaging of the ideal location. In these few cases, we imaged as close as possible to the target. Results were similar between all datasets.

Two-photon imaging

Calcium imaging procedures have been described previously. Briefly, imaging was performed with a Neurolabware 2p microscope running Scanbox 4.1 and a pulsed Ti:sapphire laser (Vision II, Coherent). Depth stability of the imaging plane was maintained using a custom plugin that made automatic movements of the objective up to every ~10s based on comparing acquired red-channel data with an initially-acquired image stack. Offline, images were run through Suite2p to perform motion correction, region-of-interest (ROI) detection and fluorescence extraction. ROIs were manually curated using the Suite2p GUI. As previously described (Zhu et al. 2022 [DOI](#)), fluorescence was neuropil-corrected, detrended, and passed through OASIS using the ‘thresholded’ method, AR1 event model, and limiting the tau parameter to be between 300 and 800 ms. Neurons were discarded if they did not meet a minimum SNR criterion. To put events on a more useful scaling, for each ROI, we found the distribution of event sizes, smoothed the distribution (ksdensity in MATLAB, with an Epanechnikov kernel and log transform), found the peak of the smoothed distribution and divided all event sizes by this value. This rescales the peak of the distribution to have a value of unity.

When time-locking the data for visualization, modulation testing, or for decoding analyses, the deconvolved events were resampled into 10-ms bins. This resampling was performed by assigning a fraction of each event into the new 10-ms bins proportionally to how much the 10-ms bins overlapped the original ~32.2 ms frames, and taking into account the exact time each ROI centroid was sampled based on its position within the field of view (FOV). Alignment for RADICaL has been described previously (Zhu et al. 2022 [DOI](#)).

Modulation index

To test whether regions of interest (ROIs) were task responsive, we adapted the “ZETA” method from (Montijn et al. 2021 [DOI](#)) to time series data, closely related to the approach used in (Heimel et al. 2023 [DOI](#)). The goal of this procedure was to determine whether the activity of each ROI was reliably modulated over time relative to some task event, such as cue or paw lift. First, neural data for all trials were resampled to 10 ms as described in the previous section. For each ROI, we averaged activity over trials and computed its fractional cumulative sum over time bins. This cumulative sum was compared to the linear baseline that would be expected from an ROI that fired uniformly at a matched rate in the same time window. The deviation of the cumulative sum from the linear baseline (i.e., the residuals) was computed, then this deviation trace was mean centered and its maximum absolute value was found. This value was the data maximum deviation. We then generated a null distribution for this statistic. To do so, we shuffled by circularly permuting the event-locked data independently for each trial before averaging across trials, mean-centering, and computing the shuffle’s maximum deviation as above. This procedure was repeated 100 times to produce the distribution for the maximum deviation under the null hypothesis that the cell fired uniformly over time. We then computed the p-value as the probability of observing the data maximum deviation under a Gumbel distribution parameterized by the mean and variance of the distribution of maximum deviations over shuffles. Compared to the ZETA procedure in (Heimel et al. 2023 [DOI](#)), our procedure may be slightly less sensitive for small time windows because of the disruption of temporal correlation at the circular permutation wraparound point. However, it uses data only from within the time window of interest, so is less sensitive to unrelated modulation due to events occurring just outside the time window of interest; and it is much faster to compute. We performed this procedure for cue-locked (–200 ms to +400 ms), lift-locked (–200 ms to +400 ms), and first-contact-locked (–300 ms to 300 ms) data, for left and right trials separately. P-values obtained through this procedure were then Bonferroni corrected for dual tests when measuring the number of cells modulated to a given event and corrected for six tests (2 targets and 3 events) when measuring the overall number of modulated

cells. Note that we did not correct for the number of ROIs tested for two reasons. First, the goal of this testing was to serve as a criterion for inclusion in subsequent decoding analyses, not to determine whether any neurons in the area at all were modulated; and second, correcting for the number of ROIs would bias comparison between areas if different numbers of ROIs were recorded in one area vs. the other.

RADICaL modeling

RADICaL models were fit using the same procedures developed in (Zhu et al. 2022 [DOI](#)) except we trained these models using the NeuroCAAS implementation (Abe et al. 2022 [DOI](#)). As in (Zhu et al. 2022 [DOI](#)), we used lift-locked data binned at 10 ms in a window from -100 to +800 ms around lift time with 100 ms padding. After models were fit, the padding was removed and RADICaL-inferred rates for each ROI were handled identically to sub-frame aligned smoothed and deconvolved data. Note that all results were qualitatively similar without RADICaL, but as expected decoding quality and correlations were generally somewhat lower.

Trial grouping

We observed considerable heterogeneity in the time from lift to spout contact (movement time) for both conditions for all mice. We reasoned that this could follow from single trial heterogeneity in either reach shape or accuracy. We therefore separated trials within each condition into two groups, determined by whether their movement time was smaller or greater than the median movement time for that condition. This procedure yielded 4 groups of trials, “brief” and “long” for left and right conditions.

Peri-event time histograms

PETHs were computed for three different locking events: cue, lift and first contact. Neural data for all trials were binned at 10 ms and smoothed with a Gaussian (35 ms s.d.). Cue and lift-locked PETHs were locked in a window of -100 to 700 ms relative to the corresponding event. Contact-locked PETHs were locked in a window of -400 to 400 ms relative to first contact. Trials were then averaged within the groups described above.

Joint angle and velocity decoding

Joint angle or velocity kinematics were linearly interpolated from their original 6.66 ms to 10 ms and smoothed with a Gaussian (15 ms s.d.). These angular variables were then treated linearly in decoding analyses as their ranges were relatively constrained during the reaching and grasping movements; although the true relationships are likely nonlinear, this serves as a sufficient approximation to demonstrate the presence of a relationship between neural activity and kinematics. We then screened for modulated neurons. ZETA p-values were sorted ascending and the top 100 cells were included in regression models in order to use equal numbers of neurons across areas. For decoding, neural and kinematic data were locked to -100 ms before and 400 ms lift onset. Deconvolved data were smoothed with a Gaussian (35 ms s.d.); RADICaL-inferred rates were not smoothed. We did not include a lag between neural activity and kinematics for simplicity; tests on selected sessions indicated that adding lags did not improve decoding. We applied PCA to the neural activity and reduced its dimensionality to be 90% of the number of included neurons. The decoder was trained and tested using cross-validated ridge regression. We partitioned the trials into 10 non-overlapping training (90%) and test (10%) set splits and fit the regularization parameter on the training set using a heuristic described in (Karabatsos 2018 [DOI](#)) with code developed in (Musall et al. 2019 [DOI](#)). We then applied the trained decoder to the test set and computed the fraction variance explained for each joint angle or velocity independently. We then repeated this same procedure for four additional 10-fold partitions for a total of 50 cross-validated folds. We report the variance explained across joints for all folds. To visualize reconstructions of the joint angles using the fit decoders, for each trial we “bagged” the decoder weights: we averaged across the 5 cross-validation folds where that trial was in the test set. This produced a denoised decoder that was used to predict the joint kinematics for that trial.

Target decoding analysis

We used logistic regression to classify individual time points as belonging to rightward or leftward targets from population activity. Neurons with significant modulation at a ZETA $p < 0.01$ threshold for any locking event and either target were included. Neural data from each trial were binned at 10 ms, locked to a window of -100 ms to 500 ms centered on cue onset, and smoothed with a Gaussian (35 ms s.d.). We partitioned the trials into five non-overlapping training (80%) and test (20%) set splits, stratifying left and right trials. We then trained a logistic regression classifier using the MATLAB *fitclinear* function on all post-cue time points across training set trials. We then projected individual trials onto the classifier dimension. We normalized each trial's projection according to the 90th percentile value over all time points of the same trial type. We report the classifier performance on all time points across test set trials. Our procedure for computing the threshold crossing times for individual trials was similar to the one used in (Kaufman et al. 2015 [DOI](#)). We computed the threshold value by first taking the median of all projections across trials at each time point, finding the maximum and minimum values of this trace, then computing their midpoint; this procedure was done separately for left and right trials. We found the threshold crossing time as the first time bin after which the projection stayed above the threshold for at least five consecutive bins (50 ms). If trials never crossed the threshold, or did not stay above threshold for more than five bins, they were discarded. We then interpolated between 10 ms bins to find the exact time that the projection crossed the threshold.

Time-generalized target decoding analysis

To assess the stability of the target-classifier dimension over time, we employed a similar logistic regression approach as above. Neural data from each trial were first interpolated as usual to 10 ms bins from -100 to 1000 ms relative to the cue, then smoothed with a Gaussian (35 ms s.d.). Time bins were then grouped into 11 non-overlapping 100 ms epochs. For each epoch, logistic regression was used to classify individual 10 ms time bins as belonging to left or right trials. A separate classifier was trained for each epoch, and thus 11 models were fit in total. This procedure was performed using 5-fold cross validation, with left and right trials stratified. The trained models for each epoch were used to predict the target identity on time points across all epochs for the held out 20% of trials from each fold. We reported the mean performance of the classifier over folds for each prediction epoch.

Reaction time decoding analysis

Deconvolved neural data were used to predict the lift reaction time (RT) on single trials. Our approach follows closely the procedure used in (Zhu et al. 2022 [DOI](#)) based on (Kaufman et al. 2016 [DOI](#)). A cross-validated logistic regression classifier was trained to discriminate pre-lift neural data (a bin -100 to 0 ms relative to lift onset) from post-lift neural data (a bin 0 to 100 ms relative to lift onset). Training was performed with *fitclinear* in Matlab, stratifying left and right trials. Neural data from the held-out trials was interpolated at 10 ms as usual, locked -200 to 600 ms relative to the lift, smoothed with a Gaussian (10 ms s.d.), then projected onto this dimension. The lift RT was calculated as the time that the projection crossed a threshold. To compute the threshold, we calculated the median across trials, then took the midpoint of the maximum and minimum. We summed the predicted time (which was effectively an offset from the true RT) with the ground truth lift RTs and correlated these with the ground truth lift RTs.

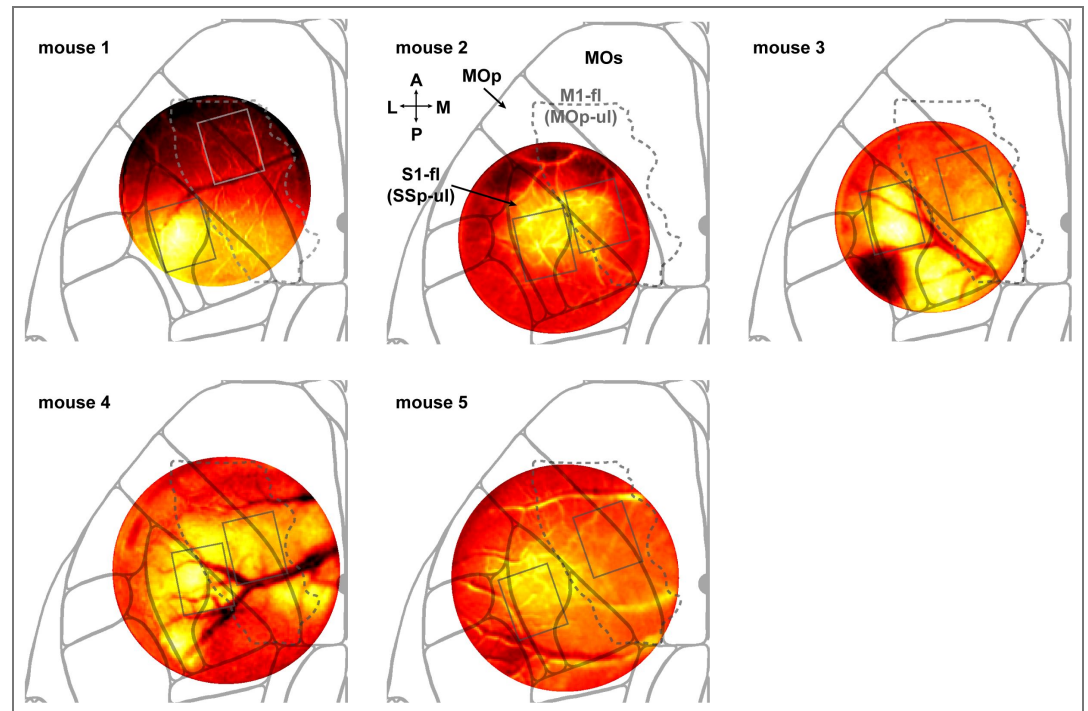


Figure 4S1. Vibration mapping for each individual mouse. Each image shows the widefield calcium response +250 ms after a vibrational stimulus applied to the contralateral paw. Each image was aligned with the underlying Allen CCF by placing the broad S1-fl (SSp-ul in the Allen nomenclature) shaped activity profile within the corresponding region. Placement was then verified against veridical injection coordinates obtained during surgery (not shown). The dashed line in each map indicates the outline of the forelimb primary motor cortex as named MOp-ul in Muñoz-Castañeda et al 2021 <https://doi.org/10.7554/eLife.106270.2> and referred to as M1-fl here. Each grey box indicates the placement of the imaging fields of view in M1-fl and S1-fl for each animal. Note that the window for mouse 1 was tilted anterior-posterior during widefield imaging, reducing the measured response in the anterior half of the window. This was accounted for in estimating atlas alignment.

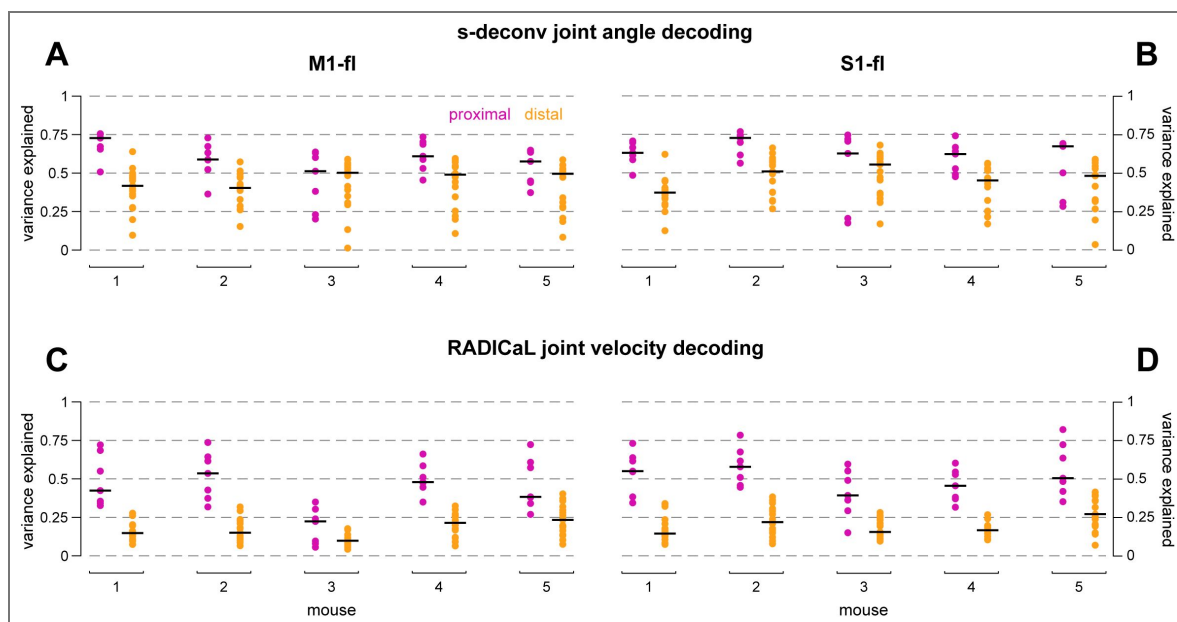


Figure 5S1.

(A) Joint angle decoding as in Figure 5, but from M1-fl deconvolved calcium transients smoothed with a 35 ms Gaussian kernel. Data is presented as in Figure 5D and 5H. (B) as in A but for S1-fl. (C) Joint velocity decoding using RADICaL rates from M1-fl. (D) Same as C but for S1-fl.

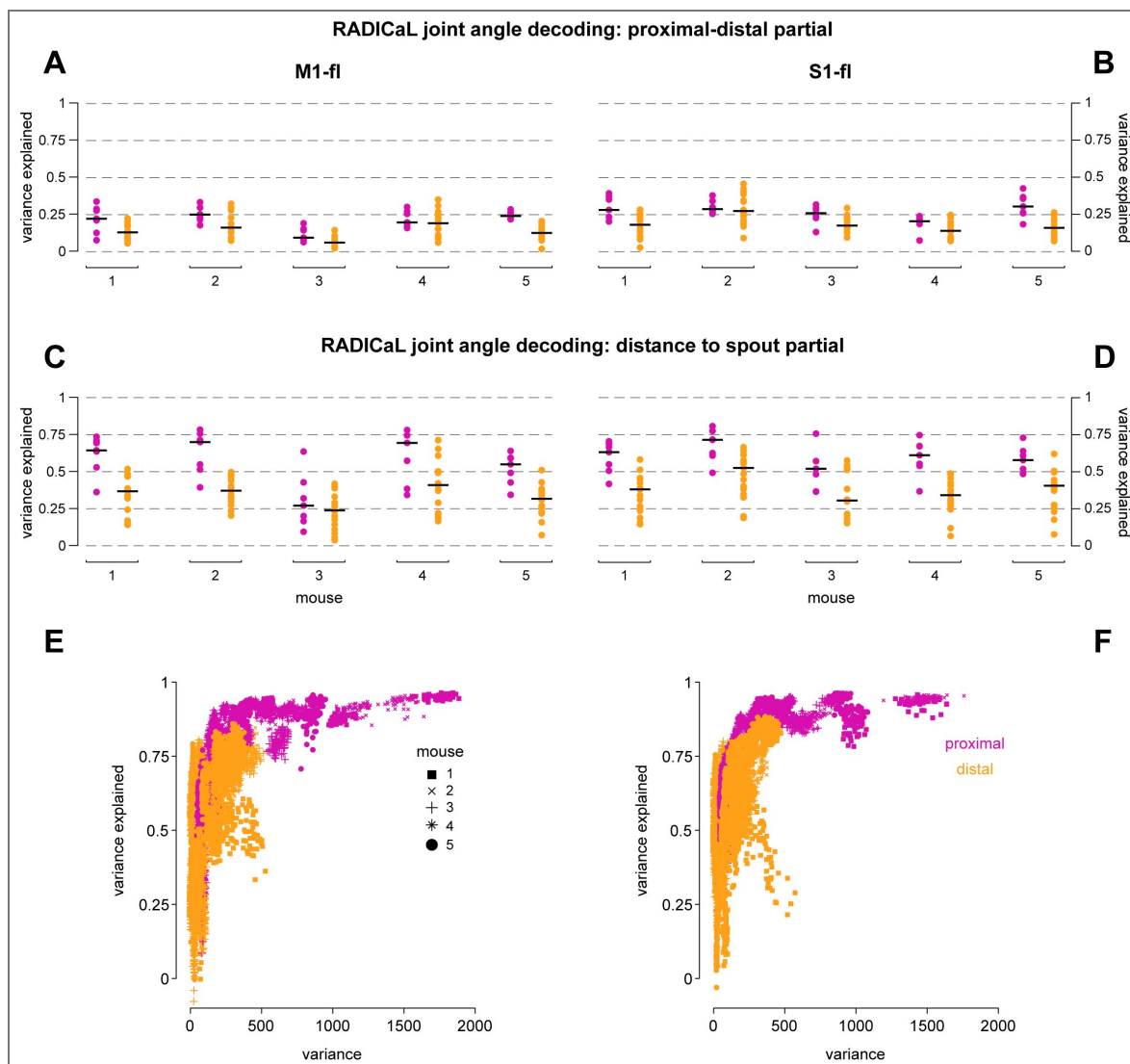


Figure 5S2.

(A) Joint angle decoding as in Figure 5 but with proximal joint angle time series regressed off of distal joints and distal joint angle time series regressed off of proximal joints. Decoding performed with RADICaL rates. Data is presented as in Figure 5D and 5H. (B) as in A but for S1-fl. (C) Joint angle decoding as in Figure 5 but with the distance to the spout of the paw centroid regressed off of all joint angles. (D) Same as C but for S1-fl. (E) Variance of joint angles and their variance explained when decoded using M1-fl RADICaL rates as in Figure 5. Each point has coordinates of the true kinematic variance and the variance explained for a single joint on a single cross-validation fold. (F) Same as E but for S1-fl.

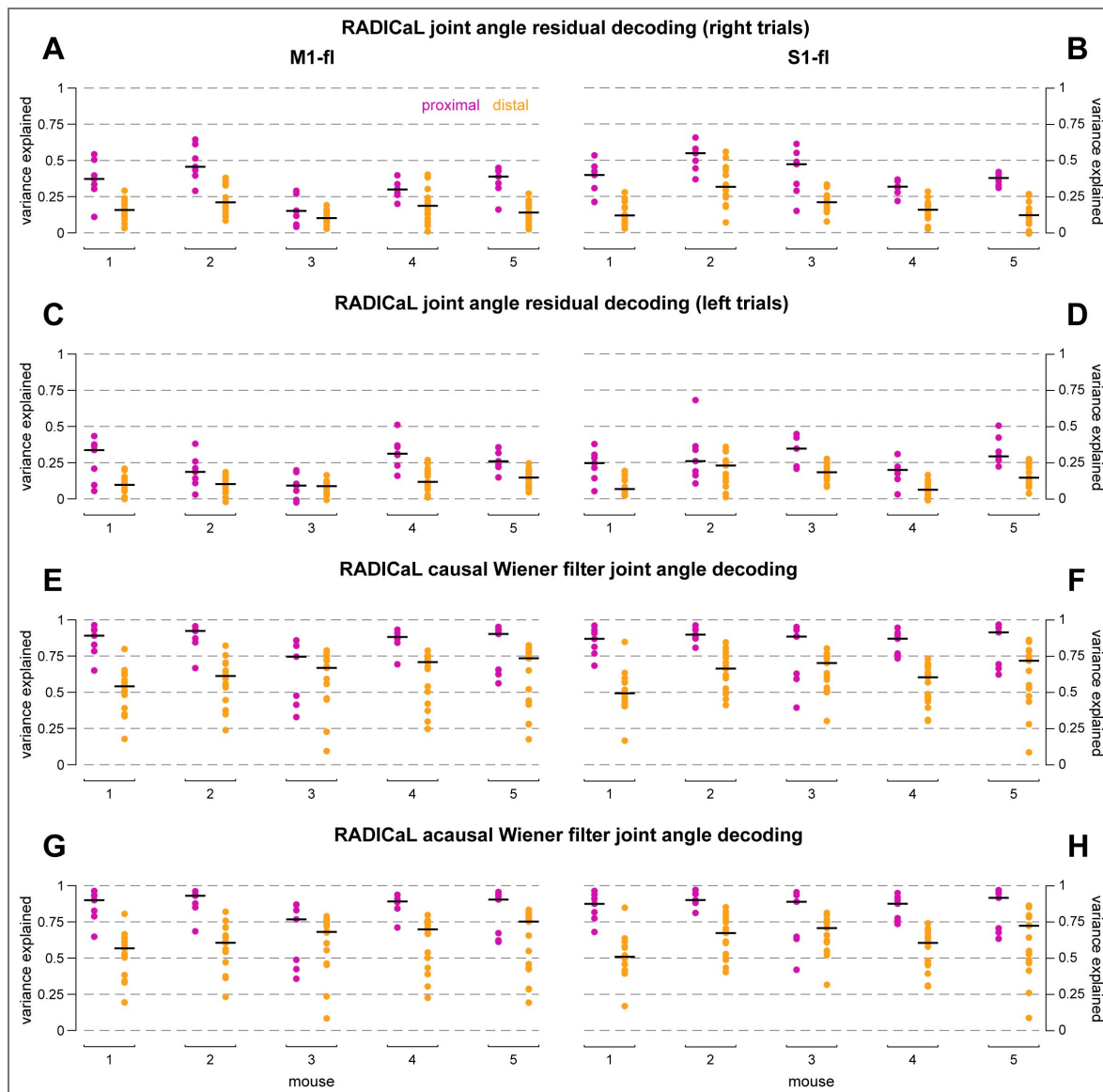


Figure 5S3.

(A) Joint angle decoding as in Figure 5S2, but for single-trial residuals obtained from subtracting off the time-varying mean over all right trials. Data is presented as in Figure 5S2. (B) Same as A but for S1-fl. (C) Joint angle decoding as in A but for left trials. (D) Same as C but for S1-fl. (E-F) Joint angle decoding as in Figure 5S2 but with additional predictors composed of causal lags at 20, 40, 60, 80 and 100 ms. (G-H) Joint angle decoding as in E-F, but including both causal and acausal lags of 20, 40, 60, 80, and 100 ms.

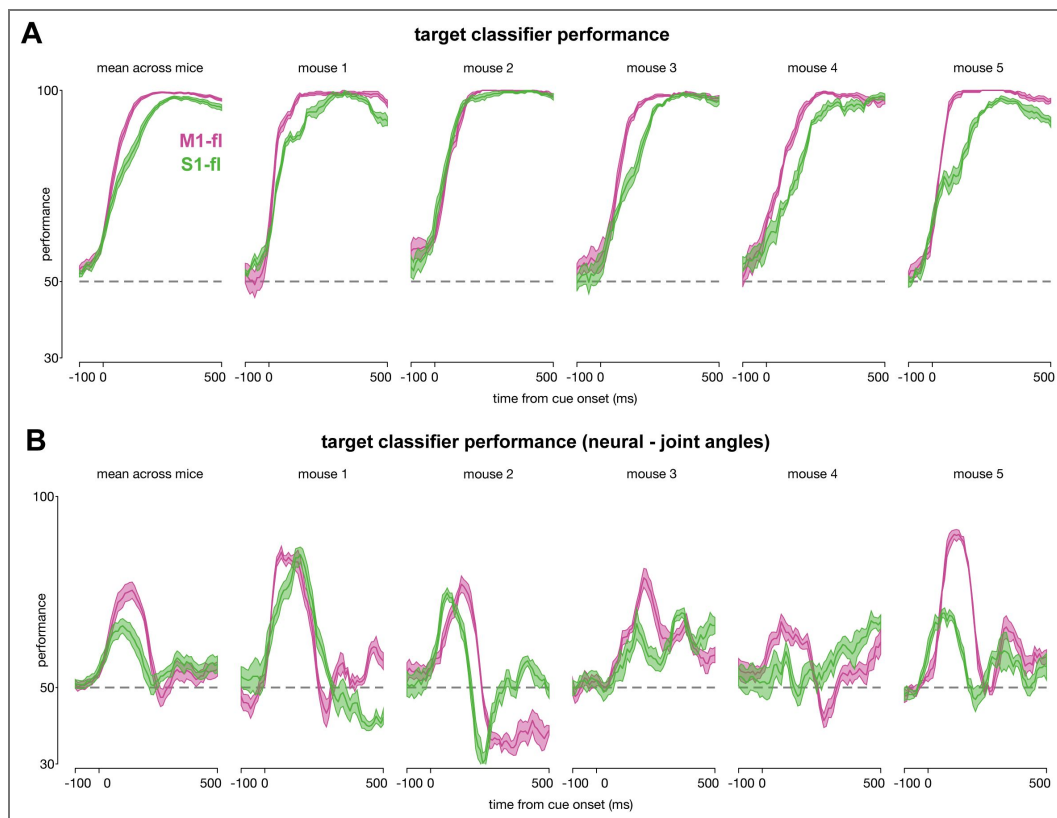


Figure 6S1.

(A) Performance over time for target classifiers from Figure 6A and 6B. Lines show mean performance across held-out trials, fills show standard error of the mean. Dashed line indicates chance performance. (B) Performance over time for target classifiers after regressing off joint angle kinematics from neural activity. Lines show mean performance across held-out trials, fills show standard error of the mean.

Data availability

Data are available on figshare. Code used to create the figures is available on GitHub: <https://github.com/kaufmanlab/GSK25-public>.

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Author contributions

H.G., Experimental design, Data acquisition, Keypoint frame labeling, Analysis and interpretation of data, Drafting and revising the manuscript; S.S. Keypoint frame labeling, Analysis and interpretation of data; M.K. Project conception, Supervision of all aspects of the project, Experimental design, Data preprocessing, Drafting and revising the manuscript, Funding.

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Peer reviews

Reviewer #1 (Public review):

Summary:

This study addresses the encoding of forelimb movement parameters using a reach-to-grasp task in mice. The authors use a modified version of the water-reaching paradigm developed by Galinanes and Huber. Two-photon calcium imaging was then performed with GCaMP6f to

measure activity across both the contralateral caudal forelimb area (CFA) and the forelimb portion of primary somatosensory cortex (fS1) as mice perform the reaching behavior. Established methods were used to extract the activity of imaged neurons in layer 2/3, including methods for deconvolving the calcium indicator's response function from fluorescence time series. Video-based limb tracking was performed to track the positions of several sites on the forelimb during reaching and extract numerous low-level (joint angle) and high-level (reach direction) parameters. The authors find substantial encoding of parameters for both the proximal and distal parts of the limb across both CFA and fS1, with individual neurons showing heterogeneous parameter encoding. Limb movement can be decoded similarly well from both CFA and fS1, though CFA activity enables decoding of reach direction earlier and for a more extended duration than fS1 activity. Collectively, these results indicate involvement of a broadly distributed sensorimotor region in mouse cortex in determining low-level features of limb movement during reach-to-grasp.

Strengths:

The technical approach is of very high quality. In particular, the decoding methods are well designed and rigorous. The use of partial correlations to distinguish correlation between cortical activity and either proximal or distal limb parameters or either low- or high-level movement parameters was very nice. The limb tracking was also of extremely high quality, and critical here to revealing the richness of distal limb movement during task performance.

The task itself also reflects an important extension of the original work by Galinanes and Huber. The demonstration of a clear, trackable grasp component in a paradigm where mice will perform hundreds of trials per day expands the experimental opportunities for the field. This is an exciting development.

The findings here are important and the support for them is solid. The work represents an important step forward toward understanding the cortical origins of limb control signals. One can imagine numerous extensions of this work to address basic questions that have not been reachable in other model systems.

Collectively, these strengths made this manuscript a pleasure to read and review.

<https://doi.org/10.7554/eLife.106270.2.sa2>

Reviewer #2 (Public review):

Summary:

In this manuscript, Grier, Salimian, and Kaufman characterize the relationship between the activity of neurons in sensorimotor cortex and forelimb kinematics in mice performing a reach-to-grasp task. First, they train animals to reach to two cued targets to retrieve water reward, measure limb motion with high resolution, and characterize the stereotyped kinematics of the shoulder, elbow, wrist, and digits. Next, they find that inactivation of the caudal forelimb motor area severely impairs coordination of the limb and prevents successful performance of the task. They then use calcium imaging to measure the activity of neurons in motor and somatosensory cortex, and demonstrate that fine details of limb kinematics can be decoded with high fidelity from this activity. Finally, they show reach direction (left vs right target) can be decoded earlier in the trial from motor than from somatosensory cortex.

Strengths:

In my opinion, this manuscript is technically outstanding and really sets a new bar for motor systems neurophysiology in the mouse. The writing and figures are clear, and the claims are supported by the data. This study is timely, as there has been a recent trend towards

recording large numbers of neurons across the brain in relatively uncontrolled tasks and inferring a widespread but coarse encoding of high-level task variables. The central finding here, that sensorimotor cortical activity reflects fine details of forelimb movement, argues against the resurgent idea of cortical equipotentiality, and in favor of a high degree of specificity in the responses of individual neurons and of the specialization of cortical areas.

Comment on revised version:

The authors addressed all my concerns, and in my opinion, the manuscript is suitable for publication of the Version of Record in its current form.

<https://doi.org/10.7554/eLife.106270.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This study addresses the encoding of forelimb movement parameters using a reach-to-grasp task in mice. The authors use a modified version of the water-reaching paradigm developed by Galinanes and Huber. Two-photon calcium imaging was then performed with GCaMP6f to measure activity across both the contralateral caudal forelimb area (CFA) and the forelimb portion of primary somatosensory cortex (fS1) as mice perform the reaching behavior. Established methods were used to extract the activity of imaged neurons in layer 2/3, including methods for deconvolving the calcium indicator's response function from fluorescence time series. Video-based limb tracking was performed to track the positions of several sites on the forelimb during reaching and extract numerous low-level (joint angle) and high-level (reach direction) parameters. The authors find substantial encoding of parameters for both the proximal and distal parts of the limb across both CFA and fS1, with individual neurons showing heterogeneous parameter encoding. Limb movement can be decoded similarly well from both CFA and fS1, though CFA activity enables decoding of reach direction earlier and for a more extended duration than fS1 activity. Collectively, these results indicate involvement of a broadly distributed sensorimotor region in mouse cortex in determining low-level features of limb movement during reach-to-grasp.

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signals. One can imagine numerous extensions of this work to address basic questions that have not been reachable in other model systems.

Collectively, these strengths made this manuscript a pleasure to read and review.

Thank you!

Weaknesses:

In the last section of the results, the authors purport to examine the representation of "higher-level target-related signals," using the decoding of reach direction. While I think the authors are careful in their phrasing here, I think they should be more explicit about what these signals could be reflecting. The "signals" here that are used to decode direction could relate to anything - low-level signals related to limb or postural muscles, or true high-level commands that dictate only what movement downstream motor centers should execute, rather than the muscle commands that dictate how. One could imagine using a partial correlation-type approach again here to extract a signal uncorrelated with all the measured low-level parameters, but there would still be all the unmeasured ones. Again, I think it is still ok to call these "high-level signals," but I think some explicit discussion of what these signals could reflect is necessary.

Thank you for this excellent suggestion. We have followed both pieces of the reviewer's advice. First, we performed the suggested analysis, partialing off the kinematics then performing target classification on the residuals. This is now Figure 6S1. The analysis revealed the presence of target-related information in the neural activity after subtracting off all linear correlations with kinematics, supporting our claims that higher-level information is present in both populations. The exact timing of classifier performances varied substantially across mice, potentially due to differences in reach-to-grasp strategy, kinematic tracking fidelity, and exact spatial locations of each recorded FOV. Following the second suggestion, we have made the relevant text more careful. We now conclude simply that higher-level signals, meaning those signals that are largely unrelated to forelimb joint angle kinematics, are present but with variable timing and strengths in each area. That text now reads:

"Target decoding performance could result from truly higher-level signals that code abstractly for target location, or alternatively could be supported by strong encoding of kinematic variables that differed between targets. To disambiguate these possibilities, we refit the linear classifier to neural data after regressing off variance related to the joint angle kinematics. The strength and exact time course of the resulting target decoding varied somewhat across animals, but the earliest portion of target decoding performance persisted in all animals after the removal of kinematics and performance remained stronger for M1-fl than S1-fl (Fig. 6S1B). We thus conclude that higher-level signals are present in both areas, but differ in their exact timing and strength. However, we note that other possible signals, such as postural changes, could not be controlled for here."

Related to this, I think the manuscript in general does not do an adequate job of explicitly raising the important caveats in interpreting parametric correlations in motor system signals, like those raised by Todorov, 2000. The authors do an expert job of handling the correlations, using PCA to extract uncorrelated components and using the partial correlation approach. However, more clarity about the range of possible signal types the recorded activity could reflect seems necessary.

This is an important point, and our text could have unintentionally misled readers. We have now attempted to make this point explicit in the Discussion and in the Results for Figure 6. This Discussion text now reads:

"Moreover, as is widely known (Todorov 2000), the exact role of these kinematically-related signals is challenging to determine from correlative measures alone; thus, determining

whether these signals are used for direct movement control or instead indirectly reflect control performed elsewhere is left as a topic for future work.”

The manuscript could also do a better job of clarifying relevant similarities and differences between the rodent and primate systems, especially given the claims about the rodent being a "first-class" system for examining the cellular and circuit basis of motor control, which I certainly agree with. Interspecies similarities and differences could be better addressed both in the Introduction, where results from both rodents and primates are intermixed (second paragraph), and in the Discussion, where more clarity on how results here agree and disagree with those from primates would be helpful. For example, the ratio of corticospinal projections targeting sensory and motor divisions of the spinal cord differs substantially between rodents and primates. As another example, the relatively high physical proximity between the typical neurons in mouse M1 and S1 compared to primates seems likely to yoke their activity together to a greater extent. There is also the relatively large extent of fS1 from which forelimb movements can be elicited through intracortical microstimulation at current levels similar to those for evoking movement from M1. All of these seem relevant in the context of findings that activity in mouse M1 and S1 are similar.

We understand two points to address here. The first point is that we needed to be more careful to attribute previous results as being from the rodent vs. monkey. We agree. We have now revised several parts of the paper to make these distinctions clearer. The second point is about the potential benefit of a thorough review of the many ways in which primate and rodent sensorimotor systems differ. We entirely agree that this could be useful for the field. However, this is a sizable endeavor and doing it full justice is beyond what we know how to fit in the space allotted for framing our results here. We therefore sought a compromise, acknowledging how our results correspond to existing results in the primate without exhaustively accounting for how they differ. Future work will be necessary to more carefully disambiguate whether species-specific differences are due to biomechanical, neurological, ethological, or as-of-yet undetermined sources. We have incorporated your final specific points about what could produce similar information in M1 and S1 into the Discussion.

“This may simply be a consequence of widely distributed representations of movement across mouse cortex (Musall et al. 2019; Steinmetz et al. 2019; Stringer et al. 2019), including forelimb somatosensory areas, or may be a consequence of the close physical proximity of M1-fl and S1-fl hindering development of functionally distinct representations (Tennant et al. 2011).”

In addition, there are a number of other issues related to the interpretation of findings here that are not adequately addressed. These are described in the Recommendations for improvement.

Reviewer #2 (Public review):

Summary:

In this manuscript, Grier, Salimian, and Kaufman characterize the relationship between the activity of neurons in sensorimotor cortex and forelimb kinematics in mice performing a reach-to-grasp task. First, they train animals to reach to two cued targets to retrieve water reward, measure limb motion with high resolution, and characterize the stereotyped kinematics of the shoulder, elbow, wrist, and digits. Next, they find that inactivation of the caudal forelimb motor area severely impairs coordination of the limb and prevents successful performance of the task. They then use calcium imaging to measure the activity of neurons in motor and somatosensory cortex, and demonstrate that fine details of limb kinematics can be decoded with high fidelity from this activity.

Finally, they show reach direction (left vs right target) can be decoded earlier in the trial from motor than from somatosensory cortex.

Strengths:

In my opinion, this manuscript is technically outstanding and really sets a new bar for motor systems neurophysiology in the mouse. The writing and figures are clear, and the claims are supported by the data. This study is timely, as there has been a recent trend towards recording large numbers of neurons across the brain in relatively uncontrolled tasks and inferring a widespread but coarse encoding of high-level task variables. The central finding here, that sensorimotor cortical activity reflects fine details of forelimb movement, argues against the resurgent idea of cortical equipotentiality, and in favor of a high degree of specificity in the responses of individual neurons and of the specialization of cortical areas.

Thank you!

Weaknesses:

It would be helpful for the authors to be more explicit about which models of mouse cortical function their results support or rule out, and how their findings break new conceptual ground.

We appreciate this feedback and have attempted to make these details clearer through changes to the Introduction and Discussion. One key change is noted below:

“The presence of detailed kinematic signals in the sensorimotor cortex supports a model of mouse sensorimotor cortex in which M1-fl and S1-fl play a strong role in shaping the fine details of reaching and grasping movements.”

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

In addition to the weaknesses noted above, I suggest the authors also address the following:

The last results section is generally lacking in statistical support for claims. Statistical support should be added.

Thank you for pointing this out, we have added more statistical support to this section.

The consideration in the Discussion of relevant previous findings and potential explanations for the distal limb signals in mouse sensorimotor cortex is somewhat lacking. There are several specific issues:

(1) In contrast to the present study, the studies cited in regards to a lack of motor cortical involvement did not involve dexterous movements - in fact, Kawai et al. explicitly engineered a task that did not involve dexterity to distinguish the role of motor cortex in learning from its known role in dextrous movement execution. In Kawai et al., the authors note one rat who adopted a more dexterous approach to the lever pressing task; in this rat, a motor cortical lesion did cause a longer-lasting reduction in task performance. In additional experiments reported in Kawai's PhD thesis, performance of a dextrous task does erode with motor cortex lesion, as seen in other studies, like the early rodent reaching work of Whishaw and colleagues.

(2) Other possible explanations for the persistence of non-dexterous tasks following motor cortical removal are compensation by, or redundant functionality in, other motor system regions.

(3) It is also worth noting that stimulation in different regions of mouse M1 and S1 evokes alternately, digit, wrist, and elbow movements in fairly similar proportions (Tennant, 2011), suggesting that descending pathways substantially target spinal circuits that control all forelimb joints.

(4) It also seems relevant that although the recovery time course is longer, nonhuman primates also retain substantial hand control after motor cortical removal (e.g. Lashley, 1925; Glees and Cole, 1950; Passingham et al., 1983). Humans of course, appear to be a different story.

These are good points. We have tried to make the Discussion better reflect the tension in the literature, including with this new text:

“However, several other previous results have indirectly suggested that M1 and S1 may be involved in the details of forelimb movement. Performance suffers with inactivation or lesioning of M1 and S1 in skilled, complex manual behaviors (Guo et al 2015, Mizes et al 2024, Whishaw et al 1990) or idiosyncratic use of digits to accomplish non-dexterous tasks (Kawai 2014). The sparing of non-dexterous tasks with these lesions may also reflect redundancy in control as opposed to irrelevance of M1 and S1. Nevertheless, our finding of low-level kinematic information in sensorimotor cortex supports a role for cortex beyond simply providing redundant high-level commands to these subcortical areas.”

We have avoided mentioning points 3 and 4 in the paper; the stimulation results might follow from activating projections not normally involved in this behavior, and discussing primates in this context would require a long list of caveats. We agree that these points are worth thinking about, but are concerned that they are too circumstantial to include in interpreting the results formally.

Although similar decoding performance is achieved using neurons from both CFA and fS1, I am left wondering whether you would do substantially better with CFA using activity at additional preceding time points, or when using exclusively time points from the past. The primary model used here appears to use neural signals from corresponding time points to decode limb parameters, but results seemingly could be different when using preceding time points as regressors.

We appreciate this suggestion and have added the analysis to an additional supplementary panel for Figure 5 (Figure 5S3). Incorporating lags into the decoder via a Wiener filter does indeed improve the decoding performance, but this could simply be due to the increase in the number of predictor variables. This analysis did not, however, further disambiguate M1-fl and S1-fl: the performance improvement was similar across areas for both causal and acausal lag configurations. This could be a consequence of the time resolution of calcium imaging, so further experiments with electrophysiology would be required to rule this possibility out. We now note this new result:

“Including additional causal (-100 ms preceding) and/or acausal (-100 ms preceding to 100 following) lags improved decoding performance modestly and similarly for both areas (Fig. 5S3E-F).”

Related to this, I am also worried about the bleeding of signals across time here. If you deconvolve and interpolate between time points, the interpolation seemingly will pull information into the past, up to half the sampling period, which here is on the order of how long it takes signals to travel to and from the limb. The authors do not make any inappropriate claims about the neural signals here reflecting causes or consequences of what is happening at the limb, but readers (like me) will still try to draw these sorts of conclusions. Is it possible that, although decoding from instantaneous signals is similar for the two regions, the M1 signals are actually motor signals related to future limb state

while the S1 signals are sensory consequences? Even if many of the relevant details related to conduction times are not known, perhaps the authors could clarify what can and can't be said related to causal interpretation here.

Thank you for suggesting further explanation here. We agree that our interpretation could be made more specific. We have added text in the Discussion section to speak more directly to what can and cannot be concluded from our analyses. In short, it is hard to be certain of lags in calcium imaging data for many reasons, and using recording methods with finer temporal resolution (like electrophysiology) will be necessary for determining the precise temporal relationships between kinematics and neural activity. In the absence of these recordings, we limit our claim to kinematic information being present in M1-fl and S1-fl neural activity and leave determining the causal role of this information to future work.

New clarifying text in the Discussion:

“The use of calcium imaging further prevents strong conclusions about whether activity reflects future limb states or sensory consequences. Confirming this limitation, inclusion of lagged data in the decoding models, whether causal or acausal, resulted in similar performance changes in both areas.”

An alternative reason why lift onset is less decodable in CFA is that CFA activates substantially before lift onset, as has been observed in previous rodent studies (Kargo and Nitz, 2004; Miri et al., 2017; Veuthey et al., 2020), perhaps as some sort of movement preparation. S1, on the other hand, may not have this early activity, and so may show a clearer transient at onset when the hand and limb start to move. This seems more likely than the explanations provided by the authors.

This is a valid possible alternative explanation and we have updated the Discussion to reflect this. This difference in the structure of M1-fl activity versus S1-fl is apparent in the projections of Figure 6A, which show M1-fl projections more clearly aligned to cue-onset than S1-fl projections.

“Our lift time decoding results are consistent with this view and align with recent observations characterizing mouse proprioceptive forelimb cortex, (Alonso et al 2023), although an alternative explanation may be simply that M1-fl activates earlier than S1-fl during reaching (Kargo and Nitz 2004; Miri et al 2017; Veuthey et al 2020).”

To better clarify relevant similarities and differences between the rodent and primate systems, the Introduction could include some of these similarities and differences exposed by the literature currently cited, and the Discussion could include an additional paragraph specifically relating findings here to previous observations in the primate.

We appreciate the reviewer’s thoughtfulness on possible framings of our results. When writing this paper, framing was a major challenge for us and we drafted quite a few versions of the Introduction including some that focused more on mouse-primate comparison. In the end, we decided the most critical function of the Intro was to set up our central question, of “levels-of-sensorimotor-control”. The rich primate literature was valuable here, but getting into a protracted compare-and-contrast exercise quickly became a distraction from the point. Further, we sought to highlight the relevance and importance of the question answered in our work as the mouse has gained prominence for filling gaps that are challenging to address with primates. This paper serves as one of many early steps towards the ultimate goal of revealing general properties of sensorimotor cortical function with the mouse model. We have made some subtle changes to the Introduction that we hope will more clearly communicate this narrative.

We agree that a Discussion paragraph directly relating our results to those in primates would benefit our conclusions and have added one:

“These results expand our understanding of the rodent sensorimotor system and highlight similarities to nonhuman primates. We show here evidence in mice of detailed joint angle kinematic signals from the full forelimb in M1 and S1, as has been shown in macaque cortex during tasks involving reaching and grasping objects (Vargas-Irwin et al. 2010; Saleh et al. 2010, 2012; Goodman et al. 2019; Okorokova et al. 2020). Additionally, the earlier onset of movement-related activity in M1-fl compared to S1-fl is similar to macaque M1 and S1 (Tanji and Evarts 1976). Taken together these results suggest that the mouse can be employed to address questions traditionally explored in primates about how cortical activity encodes detailed movement commands.”

Although this is outside the scope of the present study, it would be interesting to image descending projection neurons to see what signals are conveyed downstream, and to what targets. Some signals observed in layer 2/3 may not be strongly reflected in descending projections.

We agree that recording from descending projection neurons in this task would be of deep interest – and also agree that these experiments are beyond the scope of the present study. We look forward to performing these additional experiments in future work.

Minor:

(1) The use of “CFA” and “fS1” is a bit confusing. S1, like M1, is defined primarily based on histological criteria, while CFA is defined by intracortical microstimulation. CFA contains a substantial fraction of fS1, seemingly most of it based on the maps shown in Tennant et al., 2011. This is not really a criticism, as the field has not reached any sort of consensus on this nomenclature yet.

We are similarly unhappy with the inconsistency of the terminology in the field, and struggled with how not to make it worse. After much debate and consultation with colleagues, we decided to use “M1” and “S1” to evoke the century of literature on these areas; and “-fl” to indicate forelimb because it is more intuitive than “-ul” and avoids using the illegible “-ll” for hindlimb (relevant to our subsequent paper). For what we called M1-fl, we recorded where we did because anecdotally we saw similar responses across that swath; but note that this definition is also consistent with the definition of “MOp-ul” found with multimodal mapping by

Munoz-Castaneda (2021), which extends a little anteriorly of MOp as defined by the Allen CCF. As the field continues to mature, we hope future work can converge on a set of shared terms.

(2) Page 4: “Inactivations and lesions of M1 and S1 have shown that M1 is required for the execution of dexterous reach-to-grasp movements” - to me, earlier work from Whishaw and colleagues deserves to be cited here.

We appreciate the suggestion and have updated the references in this section to better reflect the prior work from Whishaw and other researchers.

(3) Page 5: “evoking sufficient trial-to-trial variability to avoid model overfitting.” - what I think the authors are referring to here is a particular kind of “overfitting,” the consequence of not exploring the full movement space, as opposed to model overfitting from issues with the model-fitting method itself. Rather than just saying overfitting, the authors could be clearer about what they are referring to.

The reviewer is right; the phenomenon we intended to refer to is not properly termed overfitting. Specifically, we meant that data with restricted range does not necessarily express global structure, and models can therefore incorrectly fit them. For example, fitting a

linear model to data including many periods of a sine wave will correctly show a zero-slope linear component, but fitting to only a portion of a single cycle will typically yield a nonzero slope. This is not overfitting, is not exactly underfitting (because the relevant structure is barely present in the data, as opposed to missed by an insufficiently powerful model), is not bias (the data are fit well), and is not even necessarily a problem (the local relationship may be what you are interested in). Yet, it does not reflect the larger structure of the data.

We do not know of a standard term for this phenomenon, so instead of dragging the reader through this tangential argument, we have tried to offer a simpler motivation for using multiple targets:

“Assessing the relationship between neural activity and the details of movement requires striking a balance between achieving repeatable behavior and evoking sufficient trial-to-trial variability to broadly sample movement space”.

| (4) Page 5: *Caudal Forelimb Area* should not be capitalized.

Obviated with the change in area nomenclature.

| (5) Page 7: “of linearly independent degrees of freedom” - for a neuroscience audience, I think it is better to explicitly mention that the resulting PCs are uncorrelated.

We agree that this section could benefit from clarification. We have attempted to provide additional nuance to indicate what the analysis was intended to test.

“Despite the strong coupling between the proximal and distal joint angles, rich variation remained in the action of different joints over time. The presence of strong correlations across joints suggested that the kinematics may be well described by a smaller number of independent degrees of freedom than the total number of recorded angles. To assess the number of linearly independent (uncorrelated) degrees of freedom amongst the 24 joint angles and velocities, we used double-cross-validated PCA (Yu et al. 2009); Methods; Fig. 3D), finding intermediate dimensionalities of 7 (median for joint angles) and 10 (velocities; Fig. 3E). This is consistent with the idea that joint angles across the limb are coordinated instead of controlled independently, and that this coordination is flexible enough over time to enable accurately performing reaching and grasping to different targets.”

| (6) Page 7: In the Results, the authors should mention what indicator is being used, the imaging frame rate, and summarize briefly how cells were defined.

Thank you for the suggestion, these details have been added to the relevant results section for clarity.

“To do so, we recorded neural activity from neurons in layer 2/3 M1-fl extending into the immediately adjacent secondary motor cortex (M2), and the forelimb region of S1 (S1-fl) using two-photon calcium imaging of GCaMP6f-expressing neurons in layer 2/3 (185-230 μm deep, imaged at 31 Hz, cells extracted with Suite2p (Pachitariu et al 2017)).”

| (7) Page 7: “corrected at $n=2$ ” - n doesn't typically refer to the number of tests, so for clarity I would say “corrected for dual tests.”

Thank you for pointing this out, we have corrected the text and added additional explanation in the methods for our approach to determining statistical significance across the targets and locking events.

“P-values obtained through the ZETA were then Bonferroni corrected for dual tests when measuring the number of cells modulated to a given event and corrected for six tests (2 targets and 3 events) when measuring the overall number of modulated cells.”

(8) Page 7: In the Results, when the decoding is introduced, it would be helpful to have a few details without having to hunt through the Methods. For example, were things regularized, how was cross-validation handled, etc?

Thank you for the suggestion, these details have been added to the relevant results section for clarity.

A simple linear regression model related the single-trial joint angles at all time points to single-trial neural activity at the corresponding moments. The model was fit with ridge regression, the ridge penalty was determined via a heuristic (Karabatsos 2018), and performance was measured on held-out trials (80/20 train/test split, 50 folds).

(9) Page 8: I think it is worth noting how much mouse reaching involves shoulder rotation as opposed to movement in other joints, as this seems very different from primates.

Thank you for pointing this out. We think this is mostly a task difference: our mice were in a quadrupedal stance, whereas monkeys are typically asked to reach from a sitting position. We now mention this in the Results.

“Reaching evoked particularly large rotation of the shoulder, likely because the mice reached from a quadrupedal position to targets on either side of the snout.”

(10) Page 8: Should provide quantification to clarify what is meant by "closely tracked."

We have updated the text to indicate that this claim was meant to be qualitative, and to more clearly highlight that the interest here is the first demonstration of the ability to reconstruct valid forelimb postures from decoded joint angles in the mouse. Quantifying the reconstruction properly would require substantially more manual data labeling, and the successful decoding itself demonstrates indirectly that the reconstructions are good enough to obtain the results of interest.

Additionally, we reconstructed the skeletal representation of the forelimb from the decoded joint angles and found that, as intended, the reconstructed postures had strong qualitative resemblance to the true postures, even of “minor” angles like cylindrical paw deformation or digit splay (Fig. 5C,G).

(11) Page 8: "Overall, these results suggest that instantaneous movement-related signals are similarly distributed across CFA and fS1." - I know we are being succinct here, but this sentence sounds like a non sequitur in the context of this paragraph - perhaps include a conclusion from the results in this paragraph first, then summarize the whole section.

Thank you for the suggestion, we have updated this text to more clearly conclude the results of this section.

Overall, these results reveal that neural activity in M1-fl and S1-fl is closely related to the kinematic details of reach-to-grasp movements. The ability to decode substantial variance in proximal and distal joints suggests that this relationship extends to the entire forelimb and the similar performance obtained from each area suggests that this information is similarly distributed across M1-fl and S1-fl.

(12) Page 10: Mention of projections from fS1 does not explicitly specify their preferential targeting of the dorsal horn, which seems relevant.

We appreciate the suggestion and have added this detail to the text.

Rodent S1-fl is known to influence interneuron populations in the spinal cord through direct and indirect projections that predominantly target the dorsal horn (Ueno et al. 2018), thus

these signals may also reflect S1-fl's important role in modulating reflex circuits to coordinate sensory feedback with movement generation (Moreno-López et al. 2016; Moreno-Lopez et al. 2021; Seki et al. 2003).

(13) Page 31: Labels on the figure indicating what blue and red stand for would be helpful.

Thank you for the suggestion, labels have been added to indicate left and right trials for Figure 5 C/F and Figure 6A.

(14) Page 32: Legend does not include panel D.

Thank you for catching this, the corresponding caption has been added.

Reviewer #2 (Recommendations for the authors):

(1) The Introduction could perhaps set the central question in starker relief. What specifically do the authors mean by high- vs low-level control? As suggested by the cited studies, this has been a fraught issue in primate work for decades, and I think a finer-grained framing of alternative hypotheses would help set up the results. For example, would better performance at decoding joint angles than paw position be evidence for lower-level control? The clarity of the Introduction might also be improved if the facts and unknowns were broken down by species throughout.

We have tried to further improve the focus of the Introduction on the central question, clarify what we mean, and make clearer in the review of the literature which species a finding comes from.

The clarifying text from the introduction is quoted below:

Extensive motor mapping experiments in rodents have revealed that activating different parts of the sensorimotor cortex evokes movements of different body parts or different kinds of movements of the same body part, as it does in primates (for review, see (Harrison and Murphy 2014)). Yet it is unclear how the topography of stimulation-evoked movements relates to the roles of these areas during volitional actions. Perturbations during behavioral tasks in mice involving forelimb lever or reaching movements have provided a coarse-level understanding of how these areas contribute during behavior. Inactivations and lesions of M1 and S1 have shown that M1 is required for the execution of dexterous reach-to-grasp movements (Guo et al. 2015; Sauerbrei et al. 2020; Galiñanes et al. 2018; Wang et al. 2017; Whishaw et al. 1991; Whishaw 2000) and that S1 is essential for adapting learned movements to external perturbations of a joystick (Mathis et al. 2017). However, spinal cord projections from mouse M1 and S1 primarily target spinal interneurons rather than directly synapsing onto motor neurons (Gu et al. 2017; Ueno et al. 2018; Wang et al. 2017), suggesting cortical activity might play a more modulatory role. Further, stimulation of brainstem nuclei alone can evoke naturalistic forelimb actions, including realistic reaching movements involving coordinated flexion and extension of the proximal and distal limb (Esposito et al. 2014; Ruder et al. 2021; Yang et al. 2023). Taken together, these results have raised the question of what role mouse M1 and S1 play in the control of goal-directed forelimb movements.

One route to answering this question involves characterizing the signals present in mouse M1 and S1 during movement. If mouse M1 and S1 were to control only high-level aspects of forelimb movements, activity should be dominated by 'abstract' signals like target location and reflect little trial-to-trial variability in reach kinematics. If instead M1 and S1 control low-level movement features then activity should correlate strongly with forelimb joint angle kinematics and their trial-to-trial variation when reaching to different targets. While the presence of high- or low-level signals in a cortical area does not necessarily imply that they

are causally responsible for these aspects of movement, characterizing what signals are present serves as a first step toward determining how these areas relate to movement.

(2) The kinematics and calcium traces appear to be highly stereotyped across trials. If the population encodes joint angles, would one expect to find correlations between the neural and kinematic residuals after subtraction of the time-varying means? Some additional analysis and/or discussion on this point would be helpful, especially as there are only two targets.

This is a great idea. As suggested, we implemented regression models on the residuals for each target in the new Figure 5S3. Figure 5S3 A and B show the performance when decoding the residuals for right trials and C and D show performance for left trials. Decoding remained well above chance, despite shrinking down due to predicting this relatively small within-target variation. This analysis supports our claims from the main regression models in Figure 5 and 5S1-2, and also suggests that movements ipsilateral to the reaching limb (contralateral to the recording hemisphere) may be better encoded than movements contralateral to the reaching limb. We have added a reference to this additional residual analysis in the final paragraph of the decoding section of the Results section:

“Finally, we tested whether the ability to decode these many joint angles was a direct consequence of inter-joint correlations, and might not be indicative of the presence of “real” information about some of these joints. To do so, we fit partial correlation models that removed correlations between proximal and distal joints, or removed correlations of the joint angles with a high-level parameter – the overall distance of the paw centroid to the spout. Despite substantially lowering the behavioral variance, in each case the residuals could still be decoded from neural activity (Fig 5S2A-D). Similar decoding performance for M1-fl and S1-fl was obtained from models fit to decode single-trial residuals separately for left and right trials (Fig 5S3A-D), indicating that trial-to-trial variations on each basic movement were decodable from these populations.”

Along similar lines, binary classification is used to characterize cue-, lift-, and contact-responsive neurons. Is it possible to exploit trial-to-trial variation in the cue-lift and lift-contact latencies to extract the time-varying marginal effects of each event (e.g., using a GLM)?

For the detection of single-cell modulations by different events, we have elected to retain our simple statistical test to determine modulation; in our experience, encoding models typically involve a surprising number of steps to get them to do what you actually intend. We leave more extensive encoding model-style analysis to future work, currently in progress.

(3) The authors mention prior studies suggesting that the control of some forelimb tasks can be gradually transferred from the cortex to the subcortical centers. Have they performed the inactivation at different time points across learning, and if so, do they have evidence for a diminishing effect over time (e.g., blocking of both initiation and coordination early in training)? In addition, the effects of motor cortex inactivation are similar to, but slightly different from, effects shown in reaching tasks in prior studies. Some additional discussion on this point would be useful.

Our inactivation experiments in this study were intended to coarsely demonstrate the involvement of mouse forelimb sensorimotor cortex in our task. We have not performed the inactivations over learning and leave such experiments to future work.

We agree that a little more clarity relating our results to previous ones was warranted. Previous studies (Guo et al. 2015 and Galinanes et al. 2018) have demonstrated inactivation impacts on similar tasks, but for thoroughness we sought to show the same for our task as it varied from the pellet and motorized water spout tasks in both training time and target configurations. Our results are strongly in line with those of Galinanes et al. 2018 which used

a fairly similar water spout target configuration. In the inactivation experiments of that paper, 3 out of 13 animals with initiation-triggered inactivations were able to initiate reaching within a time window similar to control trials. Additionally, a proportion of trials across multiple mice proceeded with little perturbation from the inactivations. This is consistent with our observation that M1-fl inactivations may either abolish movement initiation or allow movement initiation but impair task completion on a trial-by-trial and animal-to-animal basis. Further work is required to determine what factors influence these differential responses to inactivation and to determine how these effects differ across task variations (i.e., pellet vs water spout). We have added a brief description of these nuances to the text for clarity.

“These inactivations blocked the execution of the reach to grasp sequence, preventing the animal from making contact with the spout during the 3-second laser stimulation period (Fig. 1F; 86.5% control trials with contact within 3 seconds of cue, 5.1% inactivation trials with contact, $P < 10^{-191}$, Mann-Whitney U test, 2 mice, 495 stimulation trials). Interestingly, inactivation at the time of cue often did not prevent reach initiation (mouse 1: 54.7%, mouse 2: 34.2% of inactivation trials with lift within 3 seconds; 93.5%, 86.2% control trials). Yet the movement stalled once the paw and digits extended towards the spout, producing uncoordinated and unsuccessful reaching trajectories (Fig. 1I, two representative datasets). Taken together, these results support the involvement of M1-fl in the water-reaching task and suggest that the strength of inactivation effects may depend on specific task details like training time or target configuration (c.f. Galinanes et al. 2018).”

Minor points

(1) The rationale for the multiple comparisons procedure in identifying event-locked responses should be explained in more detail. If I understand correctly, the authors are not correcting for comparisons across ROIs, but instead control the family-wise error rate across brain regions and event types (dividing alpha by two or six). Why not instead control the false discovery rate across ROIs?

Thank you for pointing this out, it was confusing as written and we received a similar comment from Reviewer 1. We have fixed the wording now to make it clearer why we did this. We simply aimed to describe how many of the recorded neurons in each area were modulated by the task as a proxy for the engagement of these areas during the behavior, and to use this measure of modulation as a criterion for including the neuron in subsequent analysis. In other words, if the question had been “are any neurons in this area modulated by the task?” then correcting for the number of ROIs would be the correct method; but if the question is, “is this neuron probably modulated and therefore worth including in my decoder?” correcting for the number of ROIs will typically be much too conservative. Thus, we only sought to correct for the false discovery rate across events and targets for each ROI. We have added additional text in the methods to clarify these choices, below. Please also see response to (7) from Reviewer 1 above.

“Note that we did not correct for the number of ROIs tested for two reasons. First, the goal of this testing was to serve as a criterion for inclusion in subsequent decoding analyses, not to determine whether any neurons in the area at all were modulated; and second, correcting for the number of ROIs would bias comparison between areas if different numbers of ROIs were recorded in one area vs. the other.”

(2) It appears joint angles are treated as linear variables in the decoding analysis; is this correct? This seems reasonable as long as the range of motion is not too large, but the authors might briefly comment on the issue in the Methods.

Yes, all joint angles are treated as linear variables in the linear regression model. We observed empirically (as can be seen in Figure 3B and Figure 5B/F) that the joint angle

variables were relatively constrained to specific ranges during the task, with no angles displaying substantial wrap-around during the reaching and grasping movements. It is true that use of nonlinear decoding would almost surely improve performance further. Future work could also compare decoding of joint angles with muscle forces, which correlate and which we made no effort to distinguish here. In this work, though, the demonstration of a substantial relationship between neural activity and kinematics already tells us that fine details of movement are present in the M1 and S1-fl populations, which is a critical fact to understand these areas and was not previously known. We now comment explicitly on this, as suggested.

“Joint angle or velocity kinematics were linearly interpolated from their original 6.66 ms to 10 ms and smoothed with a Gaussian (15 ms s.d.). These angular variables were then treated linearly in decoding analyses as their ranges were relatively constrained during the reaching and grasping movements; although the true relationships are likely nonlinear, this serves as a sufficient approximation to demonstrate the presence of a relationship between neural activity and kinematics.”

(3) Are the limb pose estimates mirrored along the mediolateral axis? Figures 1C and 2D appear to show reaches to the left spout on the animal's right.

Thank you for pointing out the ambiguity in the display of these data. The reach trajectories were not mirrored along the mediolateral axis, but they are displayed from the perspective of the behavioral imaging cameras as shown in Figure 1A. Thus the right target reaches (ipsilateral to the animal's reaching arm) are on the left side of the camera image and the left target reaches (contralateral to the animal's reaching arm) are on the right side of the image. We have clarified this in the figure captions.

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